SMART INDIA HACKATHON 2026

BlinkNBuild OFFICIAL

OFFICIAL GOVERNMENT REPOSITORY 2026

SMART INDIA HACKATHON 2026 ALL 226 PROBLEM STATEMENTS MASTER CATALOGUE

Complete, Searchable & Categorized Offline Compendium of 172 Software and 54 Hardware Challenges Across 17 National Themes with Sponsoring Ministry Guidelines.

226	172	54	₹1 LAKH
TOTAL CHALLENGES	SOFTWARE TRACK	HARDWARE TRACK	CASH PRIZE / PS

NON-NEGOTIABLE SIH 2026 PARTICIPATION RULES

- ✓ Exactly 6 Students per Team
- ✓ Minimum 1 Female Member Mandatory
- ✓ Same College / Institute Enrolled Only
- ✓ Must Clear College SPOC Internal Round

Ministry of Education's Innovation Cell (MIC) & AICTE

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🔍 QUICK-REFERENCE MASTER INDEX (ALL 226 PS)

Sorted by PS Code & Sponsoring Ministry

PS CODE	TRACK	PROBLEM STATEMENT TITLE	THEME	SPONSORING MINISTRY
SIH26001	SOFTWARE	**AI-Based early warning and landslide Risk Monitoring System in NER**	Space Technology	Ministry of Development of North Eastern Region (MDoNER)
SIH26002	SOFTWARE	**AI-Based Smart Logistics and Accessibility Intelligence Platform for North Eastern Region (NER)**	Smart Automation	Ministry of Development of North Eastern Region (MDoNER)
SIH26003	SOFTWARE	**AI-Based Cognitive Gaming and Memory Assistance Platform for Elderly Dementia Patients in North Eastern Region (NER)**	Space Technology	Ministry of Development of North Eastern Region (MDoNER)
SIH26004	HARDWARE	**AI-Assisted Early Detection System for Osteoarthritis (OA) Risk Markers in North Eastern Region (NER)**	Space Technology	Ministry of Development of North Eastern Region (MDoNER)
SIH26005	HARDWARE	**Solar-Powered Smart Mini Cold Storage System for Fresh Vegetables in North Eastern Region (NER)**	Smart Vehicles	Ministry of Development of North Eastern Region (MDoNER)
SIH26006	SOFTWARE	**Development of an Intelligent Freight Forecasting Model for Optimized Vessel Chartering and Bulk Cargo Procurement from overseas to East Coast of India**	Smart Automation	Ministry of Steel
SIH26007	HARDWARE	**Safe and Efficient Operation of Mine Vehicles in Fog and Low-Visibility Conditions in Open Cast Iron Ore Mines.**	Smart Automation	Ministry of Steel
SIH26008	HARDWARE	**Belt Joint Rupture and Conveyor Belt Damages in Iron Ore Mining Industry: Intelligent Monitoring and Prediction of Conveyor Belt Joint Rupture and Damages in Iron Ore Mining Industry.**	Smart Automation	Ministry of Steel
SIH26009	SOFTWARE	**Using AI/ML and Space Technology to Identify Manganese Reserves and Overcome Production Shortfalls.**	Space Technology	Ministry of Steel
SIH26010	HARDWARE	**Survey/Resurvey of Rural Agricultural Land in India**	Space Technology	Ministry of Rural Development
SIH26011	SOFTWARE	**3D ULPIN Generation and vertical Property Mapping SYstem**	Space Technology	Ministry of Rural Development
SIH26012	SOFTWARE	**AI-Based Automated Urban Parcel Mapping and Cadastral Feature Extraction System using Drone Imagery**	Robotics and Drones	Ministry of Rural Development
SIH26013	SOFTWARE	**Automated Integration and Intelligent Harmonization of Multi-source Geospatial Data for urban Land Record Management.**	Disaster Management	Ministry of Rural Development
SIH26014	SOFTWARE	**An Integrated GIS-based Digital Public Infrastructure for Land Governance**	Robotics and Drones	Ministry of Rural Development

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SIH26015	SOFTWARE	**Application of Geospatial Techniques for visualization and analysis to interpret Geo-Coded Images to enhance watershed Development Outcomes.**	Disaster Management	Ministry of Rural Development
SIH26016	SOFTWARE	**Real-Time National Land Acquisition & Management System for End-to-End Digital Monitoring and Decision Support**	Agriculture, FoodTech & Rural Development	Ministry of Rural Development
SIH26017	SOFTWARE	**Predictive Analytics System for Early Detection of Land Acquisition Delays**	Agriculture, FoodTech & Rural Development	Ministry of Rural Development
SIH26018	SOFTWARE	**Intelligent Land Record Digitization and Validation System**	MedTech / BioTech / HealthTech	Ministry of Rural Development
SIH26019	SOFTWARE	**National Digital Platform for Research, Policy Innovation, and Evidence-Based Land Governance**	Blockchain & Cybersecurity	Ministry of Rural Development
SIH26020	HARDWARE	**Design and Development of Innovative Hand-Spinning Equipment for Enhancing Khadi Artisan Productivity and Income**	Blockchain & Cybersecurity	Ministry of MSME
SIH26021	SOFTWARE	**Honey Chain: A block chain-based system for honey traceability and smart beekeeping management.**	Smart Automation	Ministry of MSME
SIH26022	HARDWARE	**Design and develop a smart, solar-powered drying and compact packaging system to support home-based agarbatti manufacturing by rural women artisans.**	Agriculture, FoodTech & Rural Development	Ministry of MSME
SIH26023	SOFTWARE	**AI-Powered Geological, Mining and other Reporting Solution for CMPDI/CIL subsidiaries**	Smart Automation	Ministry of Coal
SIH26024	SOFTWARE	**AI-Based Smart Governance and Compliance Monitoring System for Coal Mines**	Smart Automation	Ministry of Coal
SIH26025	HARDWARE	**Development of an AI-enabled Low Cost Real Time Mine Subsidence Monitoring, Prediction and Early Warning System for Underground Coal Mines in India**	Disaster Management	Ministry of Coal
SIH26026	HARDWARE	**Development of Mobile (Quadruped)/Handheld Device/System for Real-Time Detection of Narcotics and Explosives across Indian Railways.**	Robotics and Drones	Ministry of Railways
SIH26027	SOFTWARE	**AI-Powered Automatic Block Planning to Maximize Asset Availability for Train Operations on Indian Railways**	Smart Automation	Ministry of Railways
SIH26028	SOFTWARE	**Dynamic Forecast of Expected Time of Arrival (ETA) for Coaching Trains**	Disaster Management	Ministry of Railways
SIH26029	HARDWARE	**Automated High-Current Short-Circuit Test System for IEC 60898-1:2015 MCB Compliance.**	Disaster Management	Ministry of Consumer Affairs, Food & Public Distribution
SIH26030	HARDWARE	**Automated Cable Specimen Preparation System for IS 10810 and IS 7098 Compliance.**	Smart Automation	Ministry of Consumer Affairs, Food & Public Distribution
SIH26031	SOFTWARE	**Quality assessment and grading of onions are often subjective and vary across procurement centers, resulting in disputes and inconsistencies.**	Fitness & Sports	Ministry of Consumer Affairs, Food & Public Distribution

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SIH26032	SOFTWARE	**Farmers often face long waiting times, lack of information regarding procurement schedules, and uncertainty about procurement status.**	Smart Automation	Ministry of Consumer Affairs, Food & Public Distribution
SIH26033	SOFTWARE	**Multiple intermediaries reduce farmers earnings and increase consumer prices.**	MedTech / BioTech / HealthTech	Ministry of Consumer Affairs, Food & Public Distribution
SIH26034	SOFTWARE	**Software System to check compliance of Packaged Commodities under Legal Metrology(Packaged Commodities) Rules, 2011 by scanning products, images and labels.**	Agriculture, FoodTech & Rural Development	Ministry of Consumer Affairs, Food & Public Distribution
SIH26035	SOFTWARE	**Development of a Software Program/Application for Generation of Test Reports for Non-Automatic Weighing Instruments (NAWI) as per OIML Recommendation R- 76**	MedTech / BioTech / HealthTech	Ministry of Consumer Affairs, Food & Public Distribution
SIH26036	SOFTWARE	**Development of an Online Verification System for Weighing and Measuring Instruments**	Smart Automation	Ministry of Consumer Affairs, Food & Public Distribution
SIH26037	SOFTWARE	**Adaptive Path Planning and Collision Avoidance for Autonomous Vehicles on Unstructured Indian Roads**	Robotics and Drones	MathWorks
SIH26038	SOFTWARE	**Explainable AI for Diabetic Retinopathy Screening in Rural India**	Clean & Green Technology	MathWorks
SIH26039	HARDWARE	**AI-Powered Underground Mine Safety, Monitoring and Rescue System.**	Travel & Tourism	Governmcnt of Jharkhand
SIH26040	HARDWARE	**Smart Water Purification and Quality Monitoring System for Rural and Mining-Affected Areas.**	Smart Automation	Governmcnt of Jharkhand
SIH26041	SOFTWARE	**AR-Based Vocational Training Simulator for Industrial Safety in Jharkhand's Mining & Manufacturing Sector**	Blockchain & Cybersecurity	Governmcnt of Jharkhand
SIH26042	SOFTWARE	**AI-Powered Vernacular Pedagogy and Real-Time Translation Tool for Mother Tongue-Based Primary Education**	Smart Education	Governmcnt of Jharkhand
SIH26043	SOFTWARE	**A digital platform to crowdsource societal challenges and facilitate collaborative problem solving through universities and industry partnerships**	MedTech / BioTech / HealthTech	Governmcnt of Jharkhand
SIH26044	SOFTWARE	**Portal for Academia - Industry collaboration for Skill Mapping, Internships and Placement**	Smart Automation	Ministry of Ayush
SIH26045	SOFTWARE	**IP-SAKTI Sahayak a multilingual, RAG-based (source-cited) AI assistant for Intellectual Property and regulatory guidance in Ayurveda, across national and international regimes.**	Toys & Games	Ministry of Ayush
SIH26046	SOFTWARE	**AIIA Clinical Trials Dashboard - a real-time, cloud-based, GCP-compliant Clinical Trial Management System (CTMS) for Ayurveda research, with CDISC/FHIR-interoperable data, role-based KPIs, and integrated ethics, regulatory (CTRI / NDCT Rules 2019) and pharmacovigilance tracking.**	Space Technology	Ministry of Ayush
SIH26047	SOFTWARE	**Patient Case-Taking Software**	Smart Automation	Ministry of Ayush

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SIH26048	HARDWARE	iKwath - a pod-based smart Kwatha (Kadha) maker that prepares a fresh, AFI/API-standardized decoction from coarse powder (yavaku?a c?r?a) on demand, in the shortest practical time without altering the decoctions quality or yield	Fitness & Sports	Ministry of Ayush
SIH26049	HARDWARE	Modifications to improve the reliability, efficiency,and lifespan of electrical and electronic equipment and systems in the ambient condition of subzero temperature and low pressure of High Altitude Areas(HAA) and Super High Altitude Areas (SHAA) of Ladakh region.	Smart Automation	DRDO
SIH26050	HARDWARE	High Altitude Performance Optimization and Robust Design of Anti-Drone System.	MedTech / BioTech / HealthTech	DRDO
SIH26051	SOFTWARE	Software Based Model Development for Design of Area Specific Shelter for Thermal Comfort Maintenance.	Agriculture, FoodTech & Rural Development	DRDO
SIH26052	HARDWARE	To develop an AI/ML-enabled adaptive noise cancellation (ANC) system that effectively suppresses stationary, non-stationary, and impulsive defence noises while maintaining high speech intelligibility and real-time performance on embedded hardware.	Smart Vehicles	DRDO
SIH26053	SOFTWARE	Adaptive Variable Resolution 2.5D Lidar Mapping for Dynamic Environment Perception	Smart Automation	DRDO
SIH26054	SOFTWARE	AI-Enabled Real-Time Digital Twin System for Health Monitoring, Fault Prediction and Mission Reliability Enhancement of Aero Piston Engines used in MALE UAVs.	Robotics and Drones	DRDO
SIH26055	SOFTWARE	Smart Scan strategy for Electronic Warfare	Clean & Green Technology	DRDO
SIH26056	SOFTWARE	Development of a Real-time Airfare Price Index for India through Automated Web Scraping of Airline and Online Travel Aggregator Portals for Augmentation of the Consumer Price Index (CPI).	Travel & Tourism	MoSPI
SIH26057	SOFTWARE	AI-Powered Automated Underwater Marine Debris and Anomaly Detection System using Side-Scan Sonar Imagery	Smart Automation	Ministry of Earth Sciences (MoES)
SIH26058	HARDWARE	Development of a Low-Power, Real-Time Adaptive Software-Defined Sonar Transmitter Payload for Autonomous Underwater Vehicles (AUVs)	Blockchain & Cybersecurity	Ministry of Earth Sciences (MoES)
SIH26059	SOFTWARE	AI-Enabled Antarctic Sea-Ice, Iceberg Trajectory, and Navigation Decision Support System	Smart Education	Ministry of Earth Sciences (MoES)
SIH26060	SOFTWARE	Digital Platform for efficient remote management of Indian Antarctic Research Stations	Disaster Management	Ministry of Earth Sciences (MoES)
SIH26061	SOFTWARE	AI-Driven Smart Energy Management System for Polar Research Stations	Renewable Energy	Ministry of Earth Sciences (MoES)
SIH26062	SOFTWARE	Integrated Polar Expedition Logistics and Asset Management System	Toys & Games	Ministry of Earth Sciences (MoES)
SIH26063	SOFTWARE	Integrated Polar Science Outreach, Knowledge Repository and Media Dissemination Portal	Space Technology	Ministry of Earth Sciences (MoES)

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SIH26064	HARDWARE	**Low-Cost Deployable Seafloor Metal Detection Sensor for Ocean Resource Exploration**	Smart Automation	Ministry of Earth Sciences (MoES)
SIH26065	HARDWARE	**Autonomous Low-Cost Ocean Observation Platform for Polar and Southern Oceans**	Smart Automation	Ministry of Earth Sciences (MoES)
SIH26066	SOFTWARE	**OceanEmbed - Satellite Embedding-Based Deep Learning Framework for Reconstruction of Subsurface Ocean Temperature from Surface Satellite Observations.**	Space Technology	Ministry of Earth Sciences (MoES)
SIH26067	SOFTWARE	**Develop a web-based interactive 3D visualization platform that integrates numerical ocean model outputs and in-situ observations.**	Smart Automation	Ministry of Earth Sciences (MoES)
SIH26068	SOFTWARE	**WeatherGPT: Conversational AI for Weather Forecasting, Alerts, and Climate Information**	Space Technology	Ministry of Earth Sciences (MoES)
SIH26069	SOFTWARE	**National Weather Big Data Analytics Platform**	Disaster Management	Ministry of Earth Sciences (MoES)
SIH26070	SOFTWARE	**To develop an Artificial Intelligence (AI) / Machine Learning (ML) based system for identification, classification, and prediction of different tropical cyclone patterns using multi-source satellite data.**	Smart Education	Ministry of Earth Sciences (MoES)
SIH26071	SOFTWARE	**AI/ML-Based Integrated heavy rainfall Early Warning and Inundation Prediction System using Satellite, Radar, observational Weather and numerical weather prediction model data.**	Disaster Management	Ministry of Earth Sciences (MoES)
SIH26072	SOFTWARE	**AIML based Nowcasting of thunderstorm and lightning using atmospheric observation including multiple radars, satellite, lightning and model data.**	Disaster Management	Ministry of Earth Sciences (MoES)
SIH26073	SOFTWARE	**AI/ML-Based Intelligent Anomaly Detection for Automatic Weather Stations (AWS)**	Disaster Management	Ministry of Earth Sciences (MoES)
SIH26074	SOFTWARE	**Downscaling of weather forecast from Block level to Panchayat level: Inferring high-resolution plots/ data/ information from low-resolution plot /data /information /variables for agro-meteorological advisory services.**	Disaster Management	Ministry of Earth Sciences (MoES)
SIH26075	SOFTWARE	**Participants are invited to design and develop **CAPACITY CONNECT A Digital Capacity Building and Learning Management Portal** to support organizational training, competency development, and knowledge sharing through a centralized web-based platform.**	Smart Education	Ministry of Earth Sciences (MoES)
SIH26076	SOFTWARE	**Development of personalized homepage for 'Mausam' mobile application:**	MedTech / BioTech / HealthTech	Ministry of Earth Sciences (MoES)
SIH26077	SOFTWARE	**AI-Driven Hyper-Local Early Warning System for Severe Weather Nowcasting**	Space Technology	Ministry of Earth Sciences (MoES)
SIH26078	SOFTWARE	**AI-Driven Spatio-Temporal Tracking of Extreme Weather Anomalies in Medium-Range Forecasts**	Disaster Management	Ministry of Earth Sciences (MoES)
SIH26079	SOFTWARE	**AI-Based Forecast Bust Detection for Medium-Range Weather Forecasts**	Disaster Management	Ministry of Earth Sciences (MoES)
SIH26080	SOFTWARE	**Regime-Aware AI Post-Processing of Monsoon Rainfall Forecasts**	Disaster Management	Ministry of Earth Sciences (MoES)

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SIH26081	SOFTWARE	Hybrid AINWP Multi-Model Forecast Blending System	Smart Automation	Ministry of Earth Sciences (MoES)
SIH26082	SOFTWARE	Air PollutionWeather Coupled Forecasting System (Delhi NCR Focus)	Disaster Management	Ministry of Earth Sciences (MoES)
SIH26083	SOFTWARE	Extreme Heatwave Early Warning and Human Thermal Stress Index	Disaster Management	Ministry of Earth Sciences (MoES)
SIH26084	SOFTWARE	Convective scale nowcasting for Thunderstorms, Hail & Cloudbursts (06 hr)	Disaster Management	Ministry of Earth Sciences (MoES)
SIH26085	SOFTWARE	Urban Flood Nowcasting System (Drainage and Rainfall Coupling)	Disaster Management	Ministry of Earth Sciences (MoES)
SIH26086	SOFTWARE	Hyperlocal Monsoon Onset & Break Prediction System (Block/Village Scale)	Agriculture, FoodTech & Rural Development	Ministry of Earth Sciences (MoES)
SIH26087	HARDWARE	AI-Enabled Cooperative Capacity Building, ERP & Employment Ecosystem	Smart Education	Ministry of Cooperation
SIH26088	HARDWARE	Multilingual Cooperative Governance & Legal Assistance Chatbot	Smart Automation	Ministry of Cooperation
SIH26089	SOFTWARE	Cooperative Gig Services Platform for Household & Community Services	Smart Automation	Ministry of Cooperation
SIH26090	SOFTWARE	AI-Driven Market Linkage and Smart Cataloging Mobile Application for Marginalized Artisans	Smart Automation	Ministry of Social Justice and Empowerment (MoSJE)
SIH26091	SOFTWARE	AI-Driven Hyper-Local Business Advisory and Financial Structuring Assistant for Rural Micro-Entrepreneurs	Agriculture, FoodTech & Rural Development	Ministry of Social Justice and Empowerment (MoSJE)
SIH26092	SOFTWARE	AI-Driven Scheme Matching for Marginalized Entrepreneurs	Smart Automation	Ministry of Social Justice and Empowerment (MoSJE)
SIH26093	SOFTWARE	AI-Based Real-Time Stress and Trauma Assessment Module for Victims/Complainants Accessing NHAA (14566) and Integrated Portal	Smart Automation	Ministry of Social Justice and Empowerment (MoSJE)
SIH26094	SOFTWARE	AI-Powered Dynamic Mental Health Monitoring and Distress Prediction System for Victims of Atrocities	MedTech / BioTech / HealthTech	Ministry of Social Justice and Empowerment (MoSJE)
SIH26095	SOFTWARE	Smart Real-Time Monitoring & Inspection Mobile App	Smart Automation	Ministry of Social Justice and Empowerment (MoSJE)
SIH26096	HARDWARE	Digital Heritage Archive for Memorials, Manuscripts & Ambedkar: AI-Powered Institutional Archive and Audio-Visual Knowledge Platform	Smart Automation	Ministry of Social Justice and Empowerment (MoSJE)
SIH26097	SOFTWARE	AI-Driven voice Assistant for livelihood Mapping and NSQF-Aligned Skilling Recommendations for SC Communities under GIA component of PM-AJAY	Smart Education	Ministry of Social Justice and Empowerment (MoSJE)

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SIH26098	HARDWARE	**Development of a Low-Cost Precision Guidance and Smart Electronic Fuze System for a 155 mm Artillery Shell**	Smart Automation	Ministry of Defence (MoD)
SIH26099	SOFTWARE	**AI-Driven Standardization and Harmonization of Material Codes Across CPSEs**	Smart Automation	Ministry of Petroleum & Natural Gas
SIH26100	SOFTWARE	**AI-Powered Integrated Bid Compliance Verification Platform for GeM Procurement**	Smart Automation	Ministry of Petroleum & Natural Gas
SIH26101	SOFTWARE	**Develop an AI enabled learning platform that identifies competency gaps, recommends personalized training through integration with the iGOT Karmayogi ecosystem, and capable of generating Quizzes and Multiple choice questions (MCQs) from uploaded learning materials to strengthen capacity building in India's Official Statistical System.**	Blockchain & Cybersecurity	MoSPI
SIH26102	SOFTWARE	**Development of an AI-powered system to detect anomalies, fraud, and inefficiencies in MPLAD Scheme implementation regd.**	Blockchain & Cybersecurity	MoSPI
SIH26103	SOFTWARE	**Use case on web-based integrated project-monitoring platform**	Smart Automation	MoSPI
SIH26104	SOFTWARE	**AI-Powered Real-Time Detection and Prevention of Voice Cloning Impersonation Attacks**	Blockchain & Cybersecurity	All India Council for Technical Education (AICTE)
SIH26105	SOFTWARE	**AI-Powered Continuous Cyber Risk Quantification and Investment Optimization Platform**	Blockchain & Cybersecurity	All India Council for Technical Education (AICTE)
SIH26106	SOFTWARE	**AI-Powered Email Threat Detection, GeoLocation and Forensic Intelligence Platform**	Blockchain & Cybersecurity	All India Council for Technical Education (AICTE)
SIH26107	SOFTWARE	**AI-powered Intelligent Assistant for Indian Standards and BIS Services for Industries and Consumers**	Smart Automation	Ministry of Consumer Affairs, Food & Public Distribution
SIH26108	SOFTWARE	**AI-Powered Recommendation Engine for Identifying Applicable Indian Standards for Procurement Specifications**	Smart Automation	Ministry of Consumer Affairs, Food & Public Distribution
SIH26109	HARDWARE	**AI-Based Predictive Modelling for Early Forecasting of Bovine Mastitis in Indian Dairy Farms**	Agriculture, FoodTech & Rural Development	Ministry of Fisheries, Animal Husbandry & Dairying
SIH26110	HARDWARE	**Development of a Low-Cost Light-weight Milk Chilling Can for Small-Scale Dairy Farmers**	Agriculture, FoodTech & Rural Development	Ministry of Fisheries, Animal Husbandry & Dairying
SIH26111	SOFTWARE	**Smart AI-Enabled Rapid Feed and Silage Quality Testing System for Dairy Farmers**	Agriculture, FoodTech & Rural Development	Ministry of Fisheries, Animal Husbandry & Dairying
SIH26112	HARDWARE	**Design and Develop a Modular Autonomous Mobile Robot (AMR) Platform for Smart Warehouse Automation**	Robotics and Drones	Autodesk

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SIH26113	HARDWARE	**Human augmentation technologies are transforming healthcare,rehabilitation, industrial ergonomics, assistive living, sports, and personal mobility by improving human capabilities and enhancing quality of life.**	MedTech / BioTech / HealthTech	Autodesk
SIH26114	SOFTWARE	**Smart City Site Planning using Autodesk Forma Site Design**	Smart Automation	Autodesk
SIH26115	SOFTWARE	**Design and Develop a Smart Mobile Medical-Waste Collection and Segregation System**	MedTech / BioTech / HealthTech	Autodesk
SIH26116	SOFTWARE	**Urban Mixed-Use Design Challenge-Design a centrally located mixed-use building in Autodesk Revit with commercial spaces (Ground + 1st floor) and residential units (up to 8 floors). 1 Level of Basement (Car Parking + EV Charging), Total (B+G+9)(Note: Plot size and all required dimensions may be assumed by students (in mm units).**	Smart Education	Autodesk
SIH26117	SOFTWARE	**Sovereign On-Premise Agentic AI Workbench using Open-Weight Multimodal LLMs for Confidential Industrial Work**	Smart Automation	Mangalore Refinery and Petrochemicals Limited (MRPL)
SIH26118	HARDWARE	**Passive Colorimetric H2S Exposure-Dosimeter Wristband with AI-Based Quantitative Reading**	Smart Automation	Mangalore Refinery and Petrochemicals Limited (MRPL)
SIH26119	SOFTWARE	**Indigenous GPU-Accelerated Optimization Solver (Sovereign Alternative to Express / CEPLEX)**	Smart Automation	Mangalore Refinery and Petrochemicals Limited (MRPL)
SIH26120	SOFTWARE	**Digital Twin for Well-to-Surface Optimization of Cyclic Steam Stimulation (CSS) and Sucker Rod Pump (SRP) Operations for Heavy Oil Wells of Baghewala Field.**	Smart Automation	Oil India Limited
SIH26121	SOFTWARE	**eRTMAC-NWIS (Nearby Wells Intelligence System): An AI-Powered Offset Well Knowledge and Decision Support Platform for Drilling Operations**	Smart Automation	Oil India Limited
SIH26122	SOFTWARE	**Intelligent Data Capture & Schedule-Linking Layer for Infrastructure Project Management: Real-Time Actual Progress Tracking (Planning-to-Execution Bridge)**	Smart Automation	Oil India Limited
SIH26123	SOFTWARE	**Edge-AI Based Distributed Fleet Coordination for Autonomous Mobile Robots (AMRs) in Smart Warehouses**	Robotics and Drones	Bharat Electronics Limited
SIH26124	SOFTWARE	**AI-Powered Mobile Urban Intelligence Platform Using Public Transport Fleet**	Fitness & Sports	Bharat Electronics Limited
SIH26125	SOFTWARE	**Blockchain-Based Secure Platform for Identity,Access Control, and Digital Asset Management**	Blockchain & Cybersecurity	Bharat Electronics Limited
SIH26126	SOFTWARE	**Vision Based Autonomous Navigation for Unmanned Ground Vehicle for Outdoor environment**	Agriculture, FoodTech & Rural Development	Bharat Electronics Limited
SIH26127	SOFTWARE	**City-Wide AI Engine for Multi-Camera ANPR Trajectory Tracking and Urban Traffic Analytics**	Smart Automation	Bharat Electronics Limited
SIH26128	SOFTWARE	**Efficient systems for early detection,prevention,and management of livestock diseases and animal health issues**	Agriculture, FoodTech & Rural Development	Government Of Maharashtra

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SIH26129	SOFTWARE	System integration and interoperability among government digital platforms,resulting in fragmented service delivery	Smart Automation	Government Of Maharashtra
SIH26130	SOFTWARE	Efficiency in streamlining industrial approvals,compliance processes,and access to government support services	Smart Automation	Government Of Maharashtra
SIH26131	SOFTWARE	Early detection and management of crop diseases and pest infestations	MedTech / BioTech / HealthTech	Government Of Maharashtra
SIH26132	SOFTWARE	Strengthening market linkages and price discovery for farmers	Agriculture, FoodTech & Rural Development	Government Of Maharashtra
SIH26133	SOFTWARE	Accessibility and quality of public healthcare services,particularly in rural and underserved areas	MedTech / BioTech / HealthTech	Government Of Maharashtra
SIH26134	SOFTWARE	Challenges in aligning skill development programs with industry requirements and emerging job market demands	Smart Automation	Government Of Maharashtra
SIH26135	SOFTWARE	Difficulties in tracking employment outcomes,skill gaps, and the impact of skilling initiatives	Smart Education	Government Of Maharashtra
SIH26136	SOFTWARE	Startup friendly public procurement mechanism that enables government departments to identify,pilot, procure,and scale innovative solutions from eligible startups	Blockchain & Cybersecurity	Government Of Maharashtra
SIH26137	SOFTWARE	Quantum-Inspired Intelligent Traffic Route Optimization in Transportation Systems Using Metaheuristic Optimization	Fitness & Sports	Egreen Quanta
SIH26138	SOFTWARE	Quantum-Inspired Fuel Consumption Prediction and Green Fleet Optimization	Smart Vehicles	Egreen Quanta
SIH26139	SOFTWARE	Hybrid Quantum Machine Learning Platform for Early Disease Detection	MedTech / BioTech / HealthTech	Egreen Quanta
SIH26140	SOFTWARE	AI-Based Interactive Quantum Algorithm Learning Platform	Smart Education	Egreen Quanta
SIH26141	SOFTWARE	Quantum-Inspired Cyber Threat Detection for Digital Signature Security	Blockchain & Cybersecurity	Egreen Quanta
SIH26142	SOFTWARE	Deep Learning Based Super Resolution Mapping (SRM) from Medium Resolution Satellite Imageries	Space Technology	National Technical Research Organisation (NTRO)
SIH26143	SOFTWARE	Leveraging satellite imagery to determine Oil spills at sea along with AIS data correlations to identify vessel responsible for the spill.	Space Technology	National Technical Research Organisation (NTRO)
SIH26144	HARDWARE	Design & Development of a High-Sensitivity Micro barometer Infrasound sensor	Smart Automation	National Technical Research Organisation (NTRO)
SIH26145	SOFTWARE	AI-Based Detection of Cyber Threats in Unidirectional IP Traffic	Blockchain & Cybersecurity	National Technical Research Organisation (NTRO)

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SIH26146	SOFTWARE	AI-Powered Monitoring & Analysis of Bitcoin Transaction Traffic	Smart Automation	National Technical Research Organisation (NTRO)
SIH26147	SOFTWARE	Automated model for analysis of .IQ and .wav files along with signal parameter extraction	Smart Automation	National Technical Research Organisation (NTRO)
SIH26148	SOFTWARE	Creation of scripts/functions with new programming language to commence Computer & Network forensic analysis without triggering security solutions	Blockchain & Cybersecurity	National Technical Research Organisation (NTRO)
SIH26149	SOFTWARE	Design and Development of an Integrated Secure Data Erasure and Advanced File Recovery Tool for Digital Forensics and Data Sanitization	Blockchain & Cybersecurity	National Technical Research Organisation (NTRO)
SIH26150	SOFTWARE	Development of a Multi-Vendor DVR/NVR Forensic Analysis Tool for Standardized Acquisition, Recovery, and Analysis of Surveillance Evidence.	Blockchain & Cybersecurity	National Technical Research Organisation (NTRO)
SIH26151	SOFTWARE	Dark web threat actor de-anonymization	Blockchain & Cybersecurity	National Technical Research Organisation (NTRO)
SIH26152	SOFTWARE	Social Media Analytics	Smart Automation	National Technical Research Organisation (NTRO)
SIH26153	SOFTWARE	AI based Network Attack Forecasting from Network Traffic Data	Space Technology	National Technical Research Organisation (NTRO)
SIH26154	SOFTWARE	Gen AI Platform for Automated Content Transformation	Smart Automation	National Technical Research Organisation (NTRO)
SIH26155	SOFTWARE	AI-Driven Multi-Vendor Network Security Compliance Auditor	Blockchain & Cybersecurity	National Technical Research Organisation (NTRO)
SIH26156	SOFTWARE	Universal Log Pre-processing Framework	Blockchain & Cybersecurity	National Technical Research Organisation (NTRO)
SIH26157	SOFTWARE	Supervisory Analytics Tool for SOC Assessment (SAT-SA)	Smart Automation	National Technical Research Organisation (NTRO)
SIH26158	SOFTWARE	Single-Pass Drone Video to Accurate 3D Model Generation System	Robotics and Drones	National Technical Research Organisation (NTRO)
SIH26159	SOFTWARE	SecureMailScope: AI-Assisted Cryptographic Security Posture Assessment for Secure Email Communications	Blockchain & Cybersecurity	National Technical Research Organisation (NTRO)
SIH26160	SOFTWARE	AI-Powered IPsec VPN Protocol Analyzer and Security Assessment Framework	Blockchain & Cybersecurity	National Technical Research Organisation (NTRO)
SIH26161	SOFTWARE	Dam Break Inundation Modelling Using Hydrodynamic Modelling of any River	Disaster Management	National Technical Research Organisation (NTRO)
SIH26162	SOFTWARE	AI-Based Detection and Classification of Industrial Fires and Persistent Thermal Sources Using NASA FIRMS, OSM & Satellite Data	Space Technology	National Technical Research Organisation (NTRO)

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SIH26163	SOFTWARE	Security Assessment of the World Monitor application	Smart Automation	National Technical Research Organisation (NTRO)
SIH26164	SOFTWARE	Enterprise Cryptographic Discovery & Analysis Tool (ECDAT)	Blockchain & Cybersecurity	National Technical Research Organisation (NTRO)
SIH26165	SOFTWARE	AI/NLP Engine to Detect Serious Injury & Fatality (SIF) Precursors in OIL's Unsafe-Act/Unsafe-Condition and Near-Miss Reports	Smart Automation	Oil India Limited
SIH26166	SOFTWARE	Multi-modal, Sun angle and scale invariant image correspondence using Chandrayaan-2 optical images (OHRC, TMC and IIRS)	Space Technology	Indian Space Research Organisation(ISRO)
SIH26167	SOFTWARE	SatQuery AI - An Interactive Vision-Language Assistant for Multimodal Remote Sensing Image Analysis through Text Queries	Space Technology	Indian Space Research Organisation(ISRO)
SIH26168	SOFTWARE	AI-ML based Intelligent Dead Reckoning system for seamless navigation	Smart Automation	Indian Space Research Organisation(ISRO)
SIH26169	SOFTWARE	Development of an AI-Based Virtual Camera Tracking System for Coarse Alignment of Mobile Free Space Optical Communication (FSOC) Terminals	Space Technology	Indian Space Research Organisation(ISRO)
SIH26170	SOFTWARE	AI-Driven Anomaly Detection in Component Burn-In & Screening	Smart Automation	Indian Space Research Organisation(ISRO)
SIH26171	SOFTWARE	On-device Visual Perception for Light-weight Browser Agents	Smart Automation	Indian Space Research Organisation(ISRO)
SIH26172	HARDWARE	Low Latency and Efficient Voice Activator for Edge Devices	Smart Automation	Indian Space Research Organisation(ISRO)
SIH26173	SOFTWARE	iTantra -Indian Multilingual TTS & STT Aided Neural Transceiver Radio Access for low bitrate links	Smart Automation	Indian Space Research Organisation(ISRO)
SIH26174	SOFTWARE	AI Human Activity Recognition for On-board BAS Experiments	Smart Automation	Indian Space Research Organisation(ISRO)
SIH26175	SOFTWARE	DepthWizard - Single-View Height Estimation and 3D Flythrough	Space Technology	Indian Space Research Organisation(ISRO)
SIH26176	SOFTWARE	ORCA Marine EcOsystem Reasoning with Collaborative Agents	Space Technology	Indian Space Research Organisation(ISRO)
SIH26177	HARDWARE	A deployable AI-powered autonomous drone that aids search-and-rescue operations by detecting people and hazards, thereby improving responder safety and reducing victim discovery time.	Disaster Management	Qualcomm Inc

PS CODE	TRACK	PROBLEM STATEMENT TITLE	THEME	SPONSORING MINISTRY
SIH26178	HARDWARE	A resilient, AI-powered environmental monitoring network that provides early detection, localized intelligence, and actionable alerts for floods, forest fires, pollution events, and other environmental hazards common in India, enabling authorities and communities to shift from reactive disaster response to proactive risk prevention.	Space Technology	Qualcomm Inc
SIH26179	HARDWARE	To build an AI-powered retail intelligence platform that delivers real-time shopper analytics, automated inventory visibility, and proactive queue management through on-device AI,enabling retailers to reduce stock-outs, improve customer experience, optimize staffing, and increase operational efficiency while maintaining privacy and minimizing cloud dependency.	Smart Automation	Qualcomm Inc
SIH26180	HARDWARE	A field-deployable AI-powered Smart Farming Assistant that helps farmers detect crop diseases, pests, nutrient deficiencies, and irrigation needs at an early stage, while improving resilience against droughts, floods, heat waves, and other agricultural risks common in India. The solution should enable higher yields, lower input costs, more efficient water usage, and faster response to emerging threats through real-time on-device intelligence.	Disaster Management	Qualcomm Inc
SIH26181	HARDWARE	A secure, AI-powered Personal Health Companion that delivers real-time, privacy-preserving health monitoring and early warning capabilities, helping individuals recognize health risks before they become emergencies. The solution should improve resilience during heat waves, floods, pollution events, and other disasters common in India while enabling continuous health support through on-device intelligence.	MedTech / BioTech / HealthTech	Qualcomm Inc
SIH26182	SOFTWARE	Automated Attribution of Unknown Cryptocurrency Wallets to Nearest Virtual Asset Service Providers (VASPs) through Blockchain Intelligence APIs	Blockchain & Cybersecurity	Ministry of Home Affairs
SIH26183	SOFTWARE	Real-Time Identification of Fraud-Linked Cryptocurrency Exchanges from Victim-Reported Suspect Wallet Addresses through Automated Blockchain Analytics	Blockchain & Cybersecurity	Ministry of Home Affairs
SIH26184	SOFTWARE	Development of a Predictive Analytics Framework for Cybercrime Complaints to Forecast Likely Cash Withdrawal Locations in Advance, Enabling Generation of Actionable Intelligence for Timely and Proactive Cybercrime Intervention.	Blockchain & Cybersecurity	Ministry of Home Affairs
SIH26185	HARDWARE	Helmet mounted conformal antenna for tactical communications in urban CQB environments.	Smart Automation	Ministry of Home Affairs
SIH26186	SOFTWARE	AI-Based Predictive Personnel Stress and Welfare Monitoring System for Uniformed Forces	MedTech / BioTech / HealthTech	Ministry of Home Affairs
SIH26187	SOFTWARE	AI-Based Intelligent Video Analytics Platform for Border Surveillance using existing CCTV Infrastructure.	Smart Automation	Ministry of Home Affairs
SIH26188	SOFTWARE	AI-Based Fake Identity & Document Screening System	Smart Automation	Ministry of Home Affairs

PS CODE	TRACK	PROBLEM STATEMENT TITLE	THEME	SPONSORING MINISTRY
SIH26189	SOFTWARE	**AI-Powered Criminal Network Analysis System**	Blockchain & Cybersecurity	Ministry of Home Affairs
SIH26190	SOFTWARE	**Secure Digital Document Management System for Legal and Investigation Documents**	Smart Automation	Ministry of Home Affairs
SIH26191	SOFTWARE	**Intelligent Identification of Hazard-Based Red Zones, Carrying Capacity Assessment, and Immediate Relocation Needs for Vulnerable Habitations**	Disaster Management	Ministry of Home Affairs
SIH26192	SOFTWARE	**Flash Flood Prediction System for Hilly Regions using Multi-Source Data Theme**	Disaster Management	Ministry of Home Affairs
SIH26193	SOFTWARE	**Student Innovation**	Smart Automation	AICTE
SIH26194	SOFTWARE	**Student Innovation**	Fitness & Sports	AICTE
SIH26195	SOFTWARE	**Student Innovation**	Smart Automation	AICTE
SIH26196	SOFTWARE	**Student Innovation**	MedTech / BioTech / HealthTech	AICTE
SIH26197	SOFTWARE	**Student Innovation**	Agriculture, FoodTech & Rural Development	AICTE
SIH26198	SOFTWARE	**Student Innovation**	Smart Automation	AICTE
SIH26199	SOFTWARE	**Student Innovation**	Fitness & Sports	AICTE
SIH26200	SOFTWARE	**Student Innovation**	MedTech / BioTech / HealthTech	AICTE
SIH26201	SOFTWARE	**Student Innovation**	Smart Automation	AICTE
SIH26202	SOFTWARE	**Student Innovation**	Smart Education	AICTE
SIH26203	SOFTWARE	**Student Innovation**	Smart Automation	AICTE
SIH26204	SOFTWARE	**Student Innovation**	Smart Education	AICTE
SIH26205	SOFTWARE	**Student Innovation**	Smart Education	AICTE
SIH26206	SOFTWARE	**Student Innovation**	Disaster Management	AICTE
SIH26207	SOFTWARE	**Student Innovation**	Travel & Tourism	AICTE
SIH26208	SOFTWARE	**Student Innovation**	Smart Education	AICTE
SIH26209	SOFTWARE	**Student Innovation**	Space Technology	AICTE
SIH26210	HARDWARE	**Student Innovation**	Smart Automation	AICTE
SIH26211	HARDWARE	**Student Innovation**	Fitness & Sports	AICTE
SIH26212	HARDWARE	**Student Innovation**	Smart Automation	AICTE

PS CODE	TRACK	PROBLEM STATEMENT TITLE	THEME	SPONSORING MINISTRY
SIH26213	HARDWARE	Student Innovation	MedTech / BioTech / HealthTech	AICTE
SIH26214	HARDWARE	Student Innovation	Agriculture, FoodTech & Rural Development	AICTE
SIH26215	HARDWARE	Student Innovation	Smart Automation	AICTE
SIH26216	HARDWARE	Student Innovation	Fitness & Sports	AICTE
SIH26217	HARDWARE	Student Innovation	MedTech / BioTech / HealthTech	AICTE
SIH26218	HARDWARE	Student Innovation	Smart Automation	AICTE
SIH26219	HARDWARE	Student Innovation	Smart Education	AICTE
SIH26220	HARDWARE	Student Innovation	Smart Automation	AICTE
SIH26221	HARDWARE	Student Innovation	Smart Education	AICTE
SIH26222	HARDWARE	Student Innovation	Smart Education	AICTE
SIH26223	HARDWARE	Student Innovation	Disaster Management	AICTE
SIH26224	HARDWARE	Student Innovation	Travel & Tourism	AICTE
SIH26225	HARDWARE	Student Innovation	Smart Education	AICTE
SIH26226	HARDWARE	Student Innovation	Space Technology	AICTE

📚 THEME-WISE DETAILED PROBLEM SPECIFICATIONS (1 TO 226)

Full Technical Scope, Input Feeds & Expected Architecture

🎯 THEME: SMART AUTOMATION

TOTAL: 74 SW: 54 HW: 20

SIH26002 SOFTWARE

Ministry of Development of North Eastern Region (MDoNER)

AI-Based Smart Logistics and Accessibility Intelligence Platform for North Eastern Region (NER)

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AI-Based Smart Logistics and Accessibility Intelligence Platform for North Eastern Region (NER). Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26006 SOFTWARE

Ministry of Steel

Development of an Intelligent Freight Forecasting Model for Optimized Vessel Chartering and Bulk Cargo Procurement from overseas to East Coast of India

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Development of an Intelligent Freight Forecasting Model for Optimized Vessel Chartering and Bulk Cargo Procurement from overseas to East Coast of India. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26007 HARDWARE

Ministry of Steel

Safe and Efficient Operation of Mine Vehicles in Fog and Low-Visibility Conditions in Open Cast Iron Ore Mines.

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Safe and Efficient Operation of Mine Vehicles in Fog and Low-Visibility Conditions in Open Cast Iron Ore Mines.. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26008 HARDWARE

Ministry of Steel

Belt Joint Rupture and Conveyor Belt Damages in Iron Ore Mining Industry: Intelligent Monitoring and Prediction of Conveyor Belt Joint Rupture and Damages in Iron Ore Mining Industry.

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Belt Joint Rupture and Conveyor Belt Damages in Iron Ore Mining Industry: Intelligent Monitoring and Prediction of Conveyor Belt Joint Rupture and Damages in Iron Ore Mining Industry.. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26021 SOFTWARE Ministry of MSME

Honey Chain: A block chain-based system for honey traceability and smart beekeeping management.

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Honey Chain: A block chain-based system for honey traceability and smart beekeeping management.. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26023 SOFTWARE Ministry of Coal

AI-Powered Geological, Mining and other Reporting Solution for CMPDI/CIL subsidiaries

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AI-Powered Geological, Mining and other Reporting Solution for CMPDI/CIL subsidiaries. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26024 SOFTWARE Ministry of Coal

AI-Based Smart Governance and Compliance Monitoring System for Coal Mines

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AI-Based Smart Governance and Compliance Monitoring System for Coal Mines. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26027 SOFTWARE Ministry of Railways

AI-Powered Automatic Block Planning to Maximize Asset Availability for Train Operations on Indian Railways

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AI-Powered Automatic Block Planning to Maximize Asset Availability for Train Operations on Indian Railways. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26030 HARDWARE Ministry of Consumer Affairs, Food & Public Distribution

Automated Cable Specimen Preparation System for IS 10810 and IS 7098 Compliance.

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Automated Cable Specimen Preparation System for IS 10810 and IS 7098 Compliance.. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26032 SOFTWARE Ministry of Consumer Affairs, Food & Public Distribution

Farmers often face long waiting times, lack of information regarding procurement schedules, and uncertainty about procurement status.

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Farmers often face long waiting times, lack of information regarding procurement schedules, and uncertainty about procurement status.. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26036 SOFTWARE Ministry of Consumer Affairs, Food & Public Distribution

Development of an Online Verification System for Weighing and Measuring Instruments

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Development of an Online Verification System for Weighing and Measuring Instruments. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26040 HARDWARE Governmcnt of Jharkhand

Smart Water Purification and Quality Monitoring System for Rural and Mining-Affected Areas.

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Smart Water Purification and Quality Monitoring System for Rural and Mining-Affected Areas.. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26044 SOFTWARE Ministry of Ayush

Portal for Academia - Industry collaboration for Skill Mapping, Internships and Placement

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Portal for Academia - Industry collaboration for Skill Mapping, Internships and Placement. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26047 SOFTWARE Ministry of Ayush

Patient Case-Taking Software

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Patient Case-Taking Software. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26049 HARDWARE DRDO

Modifications to improve the reliability, efficiency,and lifespan of electrical and electronic equipment and systems in the ambient condition of subzero temperature and low pressure of High Altitude Areas(HAA) and Super High Altitude Areas (SHAA) of Ladakh region.

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Modifications to improve the reliability, efficiency,and lifespan of electrical and electronic equipment and systems in the ambient condition of subzero temperature and low pressure of High Altitude Areas(HAA) and Super High Altitude Areas (SHAA) of Ladakh region.. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26053 SOFTWARE DRDO

Adaptive Variable Resolution 2.5D Lidar Mapping for Dynamic Environment Perception

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Adaptive Variable Resolution 2.5D Lidar Mapping for Dynamic Environment Perception. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26057 SOFTWARE Ministry of Earth Sciences (MoES)

AI-Powered Automated Underwater Marine Debris and Anomaly Detection System using Side-Scan Sonar Imagery

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AI-Powered Automated Underwater Marine Debris and Anomaly Detection System using Side-Scan Sonar Imagery. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26064 HARDWARE Ministry of Earth Sciences (MoES)

Low-Cost Deployable Seafloor Metal Detection Sensor for Ocean Resource Exploration

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Low-Cost Deployable Seafloor Metal Detection Sensor for Ocean Resource Exploration. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26065 HARDWARE Ministry of Earth Sciences (MoES)

Autonomous Low-Cost Ocean Observation Platform for Polar and Southern Oceans

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Autonomous Low-Cost Ocean Observation Platform for Polar and Southern Oceans. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26067 SOFTWARE Ministry of Earth Sciences (MoES)

Develop a web-based interactive 3D visualization platform that integrates numerical ocean model outputs and in-situ observations.

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Develop a web-based interactive 3D visualization platform that integrates numerical ocean model outputs and in-situ observations.. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26081 SOFTWARE Ministry of Earth Sciences (MoES)

Hybrid AINWP Multi-Model Forecast Blending System

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Hybrid AINWP Multi-Model Forecast Blending System. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26088 HARDWARE Ministry of Cooperation

Multilingual Cooperative Governance & Legal Assistance Chatbot

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Multilingual Cooperative Governance & Legal Assistance Chatbot. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26089 SOFTWARE Ministry of Cooperation

Cooperative Gig Services Platform for Household & Community Services

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Cooperative Gig Services Platform for Household & Community Services. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26090 SOFTWARE Ministry of Social Justice and Empowerment (MoSJE)

AI-Driven Market Linkage and Smart Cataloging Mobile Application for Marginalized Artisans

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AI-Driven Market Linkage and Smart Cataloging Mobile Application for Marginalized Artisans. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26092 SOFTWARE Ministry of Social Justice and Empowerment (MoSJE)

AI-Driven Scheme Matching for Marginalized Entrepreneurs

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AI-Driven Scheme Matching for Marginalized Entrepreneurs. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26093 SOFTWARE Ministry of Social Justice and Empowerment (MoSJE)

AI-Based Real-Time Stress and Trauma Assessment Module for Victims/Complainants Accessing NHAA (14566) and Integrated Portal

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AI-Based Real-Time Stress and Trauma Assessment Module for Victims/Complainants Accessing NHAA (14566) and Integrated Portal. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26095 SOFTWARE Ministry of Social Justice and Empowerment (MoSJE)

Smart Real-Time Monitoring & Inspection Mobile App

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Smart Real-Time Monitoring & Inspection Mobile App. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26096 HARDWARE Ministry of Social Justice and Empowerment (MoSJE)

Digital Heritage Archive for Memorials, Manuscripts & Ambedkar: AI-Powered Institutional Archive and Audio-Visual Knowledge Platform

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Digital Heritage Archive for Memorials, Manuscripts & Ambedkar: AI-Powered Institutional Archive and Audio-Visual Knowledge Platform. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26098 HARDWARE Ministry of Defence (MoD)

Development of a Low-Cost Precision Guidance and Smart Electronic Fuze System for a 155 mm Artillery Shell

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Development of a Low-Cost Precision Guidance and Smart Electronic Fuze System for a 155 mm Artillery Shell. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26099 SOFTWARE Ministry of Petroleum & Natural Gas

AI-Driven Standardization and Harmonization of Material Codes Across CPSEs

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AI-Driven Standardization and Harmonization of Material Codes Across CPSEs. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26100 SOFTWARE Ministry of Petroleum & Natural Gas

AI-Powered Integrated Bid Compliance Verification Platform for GeM Procurement

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AI-Powered Integrated Bid Compliance Verification Platform for GeM Procurement. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26103 SOFTWARE MoSPI

Use case on web-based integrated project-monitoring platform

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Use case on web-based integrated project-monitoring platform. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26107 SOFTWARE Ministry of Consumer Affairs, Food & Public Distribution

AI-powered Intelligent Assistant for Indian Standards and BIS Services for Industries and Consumers

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AI-powered Intelligent Assistant for Indian Standards and BIS Services for Industries and Consumers. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26108 SOFTWARE Ministry of Consumer Affairs, Food & Public Distribution

AI-Powered Recommendation Engine for Identifying Applicable Indian Standards for Procurement Specifications

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AI-Powered Recommendation Engine for Identifying Applicable Indian Standards for Procurement Specifications. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26114 SOFTWARE Autodesk

Smart City Site Planning using Autodesk Forma Site Design

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Smart City Site Planning using Autodesk Forma Site Design. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26117 SOFTWARE
Mangalore Refinery and Petrochemicals Limited (MRPL)

Sovereign On-Premise Agentic AI Workbench using Open-Weight Multimodal LLMs for Confidential Industrial Work

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Sovereign On-Premise Agentic AI Workbench using Open-Weight Multimodal LLMs for Confidential Industrial Work. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26118 HARDWARE
Mangalore Refinery and Petrochemicals Limited (MRPL)

Passive Colorimetric H2S Exposure-Dosimeter Wristband with AI-Based Quantitative Reading

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Passive Colorimetric H2S Exposure-Dosimeter Wristband with AI-Based Quantitative Reading. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26119 SOFTWARE
Mangalore Refinery and Petrochemicals Limited (MRPL)

Indigenous GPU-Accelerated Optimization Solver (Sovereign Alternative to Express / CEPLEX)

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Indigenous GPU-Accelerated Optimization Solver (Sovereign Alternative to Express / CEPLEX). Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26120 SOFTWARE
Oil India Limited

Digital Twin for Well-to-Surface Optimization of Cyclic Steam Stimulation (CSS) and Sucker Rod Pump (SRP) Operations for Heavy Oil Wells of Baghewala Field.

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Digital Twin for Well-to-Surface Optimization of Cyclic Steam Stimulation (CSS) and Sucker Rod Pump (SRP) Operations for Heavy Oil Wells of Baghewala Field.. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26121 SOFTWARE
Oil India Limited

eRTMAC-NWIS (Nearby Wells Intelligence System): An AI-Powered Offset Well Knowledge and Decision Support Platform for Drilling Operations

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: eRTMAC-NWIS (Nearby Wells Intelligence System): An AI-Powered Offset Well Knowledge and Decision Support Platform for Drilling Operations. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26122 SOFTWARE
Oil India Limited

Intelligent Data Capture & Schedule-Linking Layer for Infrastructure Project Management: Real-Time Actual Progress Tracking (Planning-to-Execution Bridge)

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Intelligent Data Capture & Schedule-Linking Layer for Infrastructure Project Management: Real-Time Actual Progress Tracking (Planning-to-Execution Bridge). Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26127 SOFTWARE
Bharat Electronics Limited

City-Wide AI Engine for Multi-Camera ANPR Trajectory Tracking and Urban Traffic Analytics

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: City-Wide AI Engine for Multi-Camera ANPR Trajectory Tracking and Urban Traffic Analytics. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26129 SOFTWARE
Government Of Maharashtra

System integration and interoperability among government digital platforms,resulting in fragmented service delivery

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: System integration and interoperability among government digital platforms,resulting in fragmented service delivery. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26130 SOFTWARE
Government Of Maharashtra

Efficiency in streamlining industrial approvals,compliance processes,and access to government support services

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Efficiency in streamlining industrial approvals,compliance processes,and access to government support services. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26134 SOFTWARE
Government Of Maharashtra

Challenges in aligning skill development programs with industry requirements and emerging job market demands

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Challenges in aligning skill development programs with industry requirements and emerging job market demands. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26144 HARDWARE National Technical Research Organisation (NTRO)

Design & Development of a High-Sensitivity Micro barometer Infrasound sensor

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Design & Development of a High-Sensitivity Micro barometer Infrasound sensor. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26146 SOFTWARE National Technical Research Organisation (NTRO)

AI-Powered Monitoring & Analysis of Bitcoin Transaction Traffic

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AI-Powered Monitoring & Analysis of Bitcoin Transaction Traffic. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26147 SOFTWARE National Technical Research Organisation (NTRO)

Automated model for analysis of .IQ and .wav files along with signal parameter extraction

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Automated model for analysis of .IQ and .wav files along with signal parameter extraction. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26152 SOFTWARE National Technical Research Organisation (NTRO)

Social Media Analytics

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Social Media Analytics. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26154 SOFTWARE National Technical Research Organisation (NTRO)

Gen AI Platform for Automated Content Transformation

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Gen AI Platform for Automated Content Transformation. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26157 SOFTWARE National Technical Research Organisation (NTRO)

Supervisory Analytics Tool for SOC Assessment (SAT-SA)

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Supervisory Analytics Tool for SOC Assessment (SAT-SA). Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26163 SOFTWARE National Technical Research Organisation (NTRO)

Security Assessment of the World Monitor application

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Security Assessment of the World Monitor application. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26165 SOFTWARE Oil India Limited

AI/NLP Engine to Detect Serious Injury & Fatality (SIF) Precursors in OIL's Unsafe-Act/Unsafe-Condition and Near-Miss Reports

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AI/NLP Engine to Detect Serious Injury & Fatality (SIF) Precursors in OIL's Unsafe-Act/Unsafe-Condition and Near-Miss Reports. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26168 SOFTWARE Indian Space Research Organisation(ISRO)

AI-ML based Intelligent Dead Reckoning system for seamless navigation

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AI-ML based Intelligent Dead Reckoning system for seamless navigation. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26170 SOFTWARE Indian Space Research Organisation(ISRO)

AI-Driven Anomaly Detection in Component Burn-In & Screening

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AI-Driven Anomaly Detection in Component Burn-In & Screening. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26171 SOFTWARE Indian Space Research Organisation(ISRO)

On-device Visual Perception for Light-weight Browser Agents

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: On-device Visual Perception for Light-weight Browser Agents. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26172 HARDWARE Indian Space Research Organisation(ISRO)

Low Latency and Efficient Voice Activator for Edge Devices

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Low Latency and Efficient Voice Activator for Edge Devices. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26173 SOFTWARE Indian Space Research Organisation(ISRO)

iTantra -Indian Multilingual TTS & STT Aided Neural Transceiver Radio Access for low bitrate links

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: iTantra -Indian Multilingual TTS & STT Aided Neural Transceiver Radio Access for low bitrate links. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26174 SOFTWARE Indian Space Research Organisation(ISRO)

AI Human Activity Recognition for On-board BAS Experiments

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AI Human Activity Recognition for On-board BAS Experiments. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26179 HARDWARE Qualcomm Inc

To build an AI-powered retail intelligence platform that delivers real-time shopper analytics, automated inventory visibility, and proactive queue management through on-device AI,enabling retailers to reduce stock-outs, improve customer experience, optimize staffing, and increase operational efficiency while maintaining privacy and minimizing cloud dependency.

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: To build an AI-powered retail intelligence platform that delivers real-time shopper analytics, automated inventory visibility, and proactive queue management through on-device AI,enabling retailers to reduce stock-outs, improve customer experience, optimize staffing, and increase operational efficiency while maintaining privacy and minimizing cloud dependency.. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26185 HARDWARE Ministry of Home Affairs

Helmet mounted conformal antenna for tactical communications in urban CQB environments.

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Helmet mounted conformal antenna for tactical communications in urban CQB environments.. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26187 SOFTWARE Ministry of Home Affairs

AI-Based Intelligent Video Analytics Platform for Border Surveillance using existing CCTV Infrastructure.

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AI-Based Intelligent Video Analytics Platform for Border Surveillance using existing CCTV Infrastructure.. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26188 SOFTWARE

Ministry of Home Affairs

AI-Based Fake Identity & Document Screening System

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AI-Based Fake Identity & Document Screening System. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26190 SOFTWARE

Ministry of Home Affairs

Secure Digital Document Management System for Legal and Investigation Documents

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Secure Digital Document Management System for Legal and Investigation Documents. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26193 SOFTWARE

AICTE

Student Innovation

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Student Innovation. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26195 SOFTWARE

AICTE

Student Innovation

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Student Innovation. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26198 SOFTWARE

AICTE

Student Innovation

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Student Innovation. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26201 SOFTWARE

AICTE

Student Innovation

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Student Innovation. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26203 SOFTWARE

AICTE

Student Innovation

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Student Innovation. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26210 HARDWARE

AICTE

Student Innovation

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Student Innovation. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26212 HARDWARE

AICTE

Student Innovation

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Student Innovation. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26215 HARDWARE

AICTE

Student Innovation

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Student Innovation. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26218 HARDWARE

AICTE

Student Innovation

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Student Innovation. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26220 HARDWARE

AICTE

Student Innovation

Theme: Smart Automation **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Student Innovation. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

🎯 THEME: BLOCKCHAIN & CYBERSECURITY

TOTAL: 26 SW: 24 HW: 2

SIH26019 SOFTWARE Ministry of Rural Development

National Digital Platform for Research, Policy Innovation, and Evidence-Based Land Governance

Theme: Blockchain & Cybersecurity **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: National Digital Platform for Research, Policy Innovation, and Evidence-Based Land Governance. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26020 HARDWARE Ministry of MSME

Design and Development of Innovative Hand-Spinning Equipment for Enhancing Khadi Artisan Productivity and Income

Theme: Blockchain & Cybersecurity **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Design and Development of Innovative Hand-Spinning Equipment for Enhancing Khadi Artisan Productivity and Income. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26041 SOFTWARE Governmcnt of Jharkhand

AR-Based Vocational Training Simulator for Industrial Safety in Jharkhand's Mining & Manufacturing Sector

Theme: Blockchain & Cybersecurity **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AR-Based Vocational Training Simulator for Industrial Safety in Jharkhand's Mining & Manufacturing Sector. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26058 HARDWARE Ministry of Earth Sciences (MoES)

Development of a Low-Power, Real-Time Adaptive Software-Defined Sonar Transmitter Payload for Autonomous Underwater Vehicles (AUVs)

Theme: Blockchain & Cybersecurity **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Development of a Low-Power, Real-Time Adaptive Software-Defined Sonar Transmitter Payload for Autonomous Underwater Vehicles (AUVs). Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26101 SOFTWARE MoSPI

Develop an AI enabled learning platform that identifies competency gaps, recommends personalized training through integration with the iGOT Karmayogi ecosystem, and capable of generating Quizzes and Multiple choice questions (MCQs) from uploaded learning materials to strengthen capacity building in India's Official Statistical System.

Theme: Blockchain & Cybersecurity **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

• Background India's statistical system is undergoing rapid technology advancement with increasing adoption of Artificial Intelligence (AI), Machine Learning (ML), Big Data Analytics, GIS, cloud computing, and modern statistical methodologies. Officials engaged in data collection, processing, analysis, dissemination, and policy support require continuous upskilling to meet evolving technological and domain-specific requirements. While the iGOT Karmayogi platform offers a vast repository of learning resources, officials often face challenges in identifying the most relevant courses aligned with their job roles, current competencies, and future skill requirements. Presently, there is no intelligent mechanism that performs comprehensive skill-gap assessment and recommends personalized learning pathways specifically for professionals working in Official Statistics. Artificial Intelligence (AI) and emerging digital technologies are rapidly transforming the way organizations operate, deliver services, and make decisions. However, many organizations face challenges such as limited AI awareness, skill gaps, inadequate technical expertise, and the absence of structured, scalable training mechanisms. Traditional training approaches often lack personalization, continuous assessment, and real-time learner support, making it difficult to meet diverse learning needs. An AI enabled Learning Management System can assess learners existing competencies through learnerâ€™s profile, identify skill gaps, and recommend personalized training through integration with the iGOT Karmayogi platform based on job roles, experience levels, and organizational requirements. Through adaptive learning modules, AI powered virtual assistants, automated assessments, and realtime feedback mechanisms, learners receive targeted training that improves engagement and learning outcomes. The LMS supports continuous capacity building by offering structured courses, hands-on exercises, virtual labs on emerging technologies such as Artificial Intelligence, Data Science, Cloud Computing, Cybersecurity, and Automation. AI driven analytics and dashboards enable organizations to monitor learner progress, evaluate training effectiveness, predict future skill requirements, and make informed decisions regarding workforce development. By integrating personalized learning, competency mapping, performance monitoring, and intelligent content delivery, the AI enabled LMS ensures effective adoption of emerging technologies and helps create a future-ready, digitally skilled workforce. • Detailed Description The proposed solution aims to develop an AI-enabled Skill Intelligence and Learning Platform that strengthens capacity building for officials engaged in India's Official Statistical System by integrating with the iGOT Karmayogi ecosystem. The platform should leverage Artificial Intelligence to assess competencies, identify skill gaps, and recommend personalized learning pathways aligned with each official's job role, responsibilities, and career progression. The system should automatically create a comprehensive competency profile for every official using information such as designation, department, job role, current assignment, educational qualifications, work experience, and previous trainings. Based on this profile, the platform should evaluate the official's existing competencies against predefined competency frameworks for Official Statistics and identify knowledge and skill gaps. The AI engine should map competencies across multiple domains, including: • Statistical Competencies: Survey Design, Sampling, National Accounts, Price Statistics, Labour Statistics, Agricultural Statistics, Industrial Statistics, SDG Indicators, Metadata Standards, and Data Quality Frameworks. • Technical Competencies: Python, R, SQL, Stata, SPSS, SAS, GIS, Data Visualization, AI/ML, Cloud Computing, APIs, and Open Data. • Digital Governance: Cybersecurity, Data Privacy, Digital Signatures, Government Cloud, and Digital Public Infrastructure. • Behavioural and Managerial Competencies: Leadership, Communication, Project Management, Ethics, Decision Making, and Change Management. Using AI techniques such as Machine Learning, Natural Language Processing (NLP), Large Language Models (LLMs), semantic search, and competency mapping, the platform should recommend personalized learning pathways from the iGOT Karmayogi course repository. Recommendations should consider the official's current competency level, previous learning history, departmental priorities, future job requirements, emerging technologies, and career progression. The platform should integrate seamlessly with iGOT Karmayogi APIs to retrieve course catalogues, recommend relevant courses, monitor enrolment and completion status, and update competency scores automatically. To support continuous learning, the solution should provide AI-powered virtual assistants for learner support, adaptive assessments, interactive learning modules, virtual laboratories, quizzes, and multilingual learning resources. The system should continuously monitor learner progress and dynamically update recommendations based on performance and newly acquired competencies. To strengthen capacity building, the platform should support AI powered Intelligent Assessment Engine capable of generating objective type questions (MCQs), and quizzes from uploaded learning materials such as documents, presentations, videos etc. It should provide instant evaluation, explanations for correct answers, and personalized feedback to reinforce learning outcomes. This feature should enable trainers to automatically create assessments and quizzes, and evaluate learner understanding, and provide instant feedback, thereby enhancing continuous learning and competency assessment. The engine should leverage Large Language Models (LLMs), Natural Language Processing (NLP), etc. A comprehensive analytics dashboard should be provided for both employees and administrators. The employee dashboard should display current competency levels, identified skill gaps, recommended learning paths, learning hours, and overall progress. The administrator dashboard should provide organization-wide insights into workforce competencies, training effectiveness, competency distribution, emerging skill requirements, and predictive analytics for future capacity-building needs. The solution should be designed as a secure, scalable, cloud-ready, and interoperable web platform capable of integrating with existing government digital ecosystems through standard APIs. The platform should support role-based access control, Single Sign-On (SSO), and secure data exchange while ensuring compliance with government cybersecurity and data privacy guidelines. The proposed AI-enabled Skill Intelligence Platform will enable data-driven workforce development by delivering personalized, competency-based learning recommendations, improving utilization of iGOT Karmayogi resources, and creating a future-ready statistical workforce equipped with modern statistical, analytical, and digital skills required for the evolving needs of India's Official Statistical System. • Expected Solution The AI enabled platform for training and capacity building by providing personalized learning recommendations, improving competency levels of officials, enhancing utilization of iGOT Karmayogi resources, and creating a future-ready workforce equipped with modern statistical and digital skills. Additionally, AI powered generation of objective type questions and quizzes from uploaded learning content for automated assessments and self evaluation. The solution should provide: • AI-based competency assessment • Automated skill-gap analysis • Seamless iGOT integration • Personalized learning recommendations of iGOT Course Module as well as NSSTAâ€™s TPAC recommended Training Programme • AI powered generation of MCQ and Quizzes from uploaded learning content. • Interactive dashboards for Learner and Administrator • Secure, and scalable web application

SIH26102 SOFTWARE MoSPI

Development of an AI-powered system to detect anomalies, fraud, and inefficiencies in MPLAD Scheme implementation regd.

Theme: Blockchain & Cybersecurity **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Development of an AI-powered system to detect anomalies, fraud, and inefficiencies in MPLAD Scheme implementation regd.. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26104 SOFTWARE		All India Council for Technical Education (AICTE)

AI-Powered Real-Time Detection and Prevention of Voice Cloning Impersonation Attacks

Theme: Blockchain & Cybersecurity **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

• Background Recent advancements in generative AI and neural speech synthesis have made high-fidelity voice cloning possible from only a few seconds of recorded audio. Threat actors are exploiting these capabilities to impersonate CXOs,government officials, and trusted individuals in order to initiate fraudulent financial transactions, manipulate employees,or bypass verification procedures in high-risk workflows. Conventional call verification methodsâ€"such as caller ID, manual call-back, and basic voice familiarityâ€"are no longer sufficient to distinguish genuine callers from AI-generated or manipulated voices, especially in high-pressure social engineering scenarios. These attacks are increasingly orchestrated over VoIP, mobile networks, and enterprise collaboration platforms,sometimes combined with leaked personal information to create highly convincing narratives. The absence of automated detection of synthetic or cloned voices in real time significantly increases the likelihood of large-scale financial fraud and reputational damage to institutions that rely heavily on telephonic instructions and approvals. • Problem Statement Current telephony and communication ecosystems lack a robust, AI-driven mechanism to detect and flag voice cloning or synthetic speech impersonation during live calls. Existing solutions rarely perform granular analysis of acoustic artifacts, prosody, and speech generation patterns and typically cannot provide an actionable risk score while the conversation is ongoing.There is a need for an end-to-end security framework that can analyze incoming voice streams in near real time,determine the likelihood that the caller is using a cloned or AI-generated voice, and provide timely alerts and recommendations before sensitive actionsâ€"such as approval of fund transfers or disclosure of confidential informationâ€"are taken. The solution must be privacy-preserving, scalable across telecom and enterprise environments, and support multilingual contexts with diverse Indian accents and dialects. • Proposed Solution Develop an AI-powered, real-time voice integrity verification framework that integrates advanced deep learning,digital signal processing, and contextual analysis to detect AI-generated or manipulated voices. The system should continuously process live or near live audio streams from telephony, VoIP, and collaboration platforms,extract discriminative features, and compute a dynamic impersonation risk score.The framework should expose APIs and SDKs for seamless integration with banking applications, enterprise communication systems, and telecom operator infrastructures, enabling proactive fraud prevention and enhanced cyber resilience in voice channels. • Key Components • Multi-Layer Voice Authenticity Analysis o Acoustic and spectral analysis using deep learning models to detect synthesis artifacts, phase inconsistencies, and spectral signatures indicative of cloned or AI-generated audio. o Prosody and behavioral analysis to model speech rhythm, pitch contours, pauses, and microvariations, differentiating natural human speech from neural TTS outputs. o Cross-session consistency checks comparing ongoing call features against historical genuine samples (where available) to detect anomalies in speaker identity. • Real-Time Risk Scoring Engine o Continuous computation of a confidence/risk score indicating the probability of impersonation or synthetic speech. o Threshold-based alerting logic configurable for different risk scenarios (for example, high-value transaction calls, privileged access approvals). o Contextual enrichment using metadata such as call origin, known contact information, transaction context, and historical fraud indicators. • Alerting and User Interaction Layer o Multi-channel alert mechanisms (UI prompts, SMS/email, in-app notifications) for frontline staff and end users. o Pre-transaction warning prompts recommending secondary verification such as call-back, multifactor authentication, or escalation to supervisors. o Configurable workflows for banks,enterprises, and government agencies to define automated responses when impersonation risk crosses thresholds. • Privacy and Compliance Module o Minimal retention of voice recordings with options for on-device or edge inference to reduce central storage of sensitive audio data. o Support for anonymization or feature-only logging to comply with data protection and privacy requirements. • Platform and Integration APIs o REST/gRPC APIs and SDKs for integration with core banking systems, contact center platforms, enterprise communication tools, and telecom networks. o Support for multiple Indian languages and regional accents through language-agnostic feature extraction and language-specific acoustic models. • Expected Outcomes • Significant reduction in financial fraud and social engineering incidents driven by voice cloning and AIenabled impersonation. • Improved trust and assurance in voice-based communication channels for individuals, financial institutions, enterprises, and government organizations. • Early detection of AI-driven social engineering attacks, enabling proactive containment and incident response. • A reusable security layer for telecom operators and enterprises that strengthens overall cyber resilience and aligns with national cybersecurity objectives.

SIH26105 SOFTWARE All India Council for Technical Education (AICTE)

AI-Powered Continuous Cyber Risk Quantification and Investment Optimization Platform

Theme: Blockchain & Cybersecurity **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

• Background Enterprises and institutions invest heavily in cybersecurity tools, compliance programs, and risk management initiatives, yet cyber risk is still predominantly communicated using qualitative ratings such as 'Low','Medium', or 'High'. These coarse categories fail to express the potential financial impact of cyber threats,making it difficult for senior management, boards, and regulators to evaluate whether current cyber investments are adequate or optimally allocated. Cyber risk is inherently dynamic: new vulnerabilities emerge, threat actors change tactics, business services are added or retired, and security controls mature over time. Most current risk assessment practices rely on periodic, manual exercises, resulting in stale risk registers and limited visibility into the organizationâ€™s real-time cyber exposure. This gap leads to suboptimal prioritization of remediation efforts, under- or over-spending on security controls, and weak alignment between technical risk metrics and business decision-making. • Problem Statement Design and develop an AI-powered platform that continuously quantifies cyber risk in monetary terms by correlating technical security telemetry with business asset criticality and control effectiveness. The platform must estimate the likelihood and financial impact of cyber incidents, identify key risk drivers, and recommend cost-effective mitigation strategies under explicit budget constraints.The solution should bridge the gap between technical cybersecurity metrics and business language, enabling CISOs, risk officers, and executive leadership to make informed, data-driven decisions about cyber risk and security investment. • Proposed Solution Develop a cloud-ready cyber risk analytics platform that ingests data from multiple enterprise security and IT sourcesâ€”such as vulnerability management, SIEM, IAM, EDR, CSPM, asset inventories, and threat intelligence feedsâ€”and uses AI/ML models to compute continuous risk scores and estimated financial exposure, such as Expected Annual Loss. The system should provide interactive dashboards and decision-support tools that allow stakeholders to simulate remediation scenarios, evaluate investment options, and understand the return on security investment.The platform must be capable of mapping risk metrics to established cybersecurity frameworks, including ISO/IEC 27001, NIST Cybersecurity Framework, CIS Controls, RBI Cyber Security Framework, and SEBI Cybersecurity and Cyber Resilience Framework, supporting both regulatory reporting and internal governance. • Key Components • Risk Quantification Engine o Continuous aggregation and normalization of data from vulnerability scanners, SIEM, IAM, EDR,CSPM, asset inventory, and other security tools. o Statistical and ML-based estimation of incident likelihood and potential business impact, including downtime costs, data breach costs, regulatory penalties, and reputational effects. o Calculation of enterprise cyber risk as financial exposure metrics (for example, Expected Annual Loss and Value at Risk) at organization, business unit, and asset levels. o Asset criticality modeling to weigh technical findings based on business importance and service dependencies. o Control effectiveness evaluation using telemetry about configuration strength,incident history, and compliance status. • AI Decision Support Layer o Predictive analytics for emerging threats and evolving risk based on trends in vulnerabilities, threat intelligence, and control performance. o AI-generated mitigation recommendations that propose prioritized actionsâ€”such as patch deployment, access control tightening, network segmentation, and additional monitoringâ€—with quantified risk reduction. o Natural language query interface for non-technical stakeholders, enabling questions like 'What is our highest financial cyber risk today?' or 'Which vulnerabilities contribute most to our expected losses?'. o Scenario simulation tools for exploring 'what-if' analyses, such as 'What happens if MFA is implemented across all privileged accounts?' or 'How will delaying remediation by 30 days affect our financial exposure?'. • Investment Optimization Module o Optimization models that recommend sets of controls and remediation actions delivering maximum risk reduction for a specified budget (for example, ?1 crore). o Computation of ROSI and cost-benefit metrics for different security initiatives to support strategic planning and board-level approvals. o Visualization of 'Investment vs. Risk Reduction' curves to highlight diminishing returns and optimal spend zones. • Executive and Technical Dashboards o Unified views for CISOs and executives, including Enterprise Risk Score, total Financial Exposure, Risk Trend Analysis, Top Risk Contributors, and Risk Reduction Opportunities. o Drill-down capability for technical teams to see control-level and asset-level findings, remediation backlogs, and mapping to frameworks and policies. • Compliance and Framework Mapping o Built-in mapping against frameworks such as ISO/IEC 27001, NIST Cybersecurity Framework, CIS Controls, RBI Cyber Security Framework, and SEBI Cybersecurity and Cyber Resilience Framework. o Support for generating evidence-based reports and dashboards for audits, regulatory filings, and internal governance committees. • Expected Outcomes • Continuous, near real-time visibility into enterprise cyber risk, expressed in monetary terms understandable to business stakeholders. • Improved prioritization of cybersecurity initiatives based on quantified impact rather than subjective risk ratings. • Enhanced communication of cyber risk to executive management, boards, and regulators through intuitive, data-driven dashboards and narratives. • More rational and optimized cybersecurity investment decisions, maximizing risk reduction per unit of spend. • Reduction in both the likelihood and financial impact of cyber incidents through targeted remediation and investment strategies.

SIH26106 SOFTWARE All India Council for Technical Education (AICTE)

AI-Powered Email Threat Detection, GeoLocation and Forensic Intelligence Platform

Theme: Blockchain & Cybersecurity **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

• Background Email continues to be one of the most widely used communication channels in government, education, banking,and enterprise ecosystems. However, it also remains one of the most exploited attack vectors for phishing,impersonation, business email compromise, financial fraud, credential theft, and malware delivery. Threat actors increasingly use spoofed domains, deceptive sender identities, social engineering techniques, and compromised infrastructure to send highly convincing fraudulent emails that appear legitimate to end users.Traditional email security controls such as spam filters, static blacklists, and rule-based signature mechanisms are often insufficient to detect sophisticated fraudulent emails. Attackers now use AI-generated language,domain lookalikes, display-name spoofing, hidden redirection links, and relay chains to evade standard detection systems. In many cases, even when a suspicious email is identified, organizations lack the technical capability to effectively trace the source path, identify probable sender infrastructure, correlate geolocation clues, and support investigation into the origin of the email.This gap creates major challenges for cybersecurity teams, law enforcement support, fraud response units, and institutional administrators who need not only to detect malicious emails but also to investigate their source and reveal indicators that may help identify the actor or infrastructure behind the attack. • Problem Statement Current email security ecosystems primarily focus on filtering or blocking suspicious content but provide limited intelligence for deep forensic tracing of fraudulent email origins. Existing tools often do not adequately correlate email headers, SMTP relay paths, SPF/DKIM/DMARC validation results, IP reputation, geolocation indicators, domain registration intelligence, and behavioral patterns to build a complete picture of the senderâ€™s identity or operating location.There is a need for an AI-powered platform capable of detecting phishing, spoofed, impersonated, and fraudulent emails in real time or near real time, analyzing the complete technical structure of an email, tracing its transmission path across mail servers, estimating its origin with location, and generating forensic intelligence and investigative insights that assist in identifying malicious infrastructure, compromised systems, or threat actors behind the attack.The solution should support forensic analysis, fraud prevention, institutional email security, and investigation workflows while maintaining legal, privacy, and evidentiary standards. • Proposed Solution Develop an AI-Powered Email Threat Detection, GeoLocation and Forensic Intelligence Platform that combines Natural Language Processing (NLP), Machine Learning (ML), email header forensics, IP intelligence, domain analysis, and graph-based correlation to identify suspicious emails,detect advanced email threats, and investigate their probable origin. The system should ingest raw email content, metadata, and headers; validate sender authentication mechanisms; extract indicators of compromise; reconstruct relay paths; analyze originating IP addresses and associated geolocation data; and generate a confidence-based assessment of fraud risk and probable sender origin. The platform should provide actionable alerts, visual trace maps, and forensic reports for security analysts,administrators, and investigators. • Key Components • Fraudulent Email Detection Engine o NLP-based analysis of email subject lines, body text, urgency cues, impersonation language, and social engineering patterns. o Detection of phishing indicators such as spoofed sender addresses, deceptive domains, suspicious attachments, malicious links, and obfuscated URLs. o AI/ML models to classify emails as legitimate, suspicious, impersonated, phishing, or fraud-related. o Identification of business email compromise patterns such as payment diversion, fake invoice requests, credential harvesting attempts, and executive impersonation. • Email Header and Protocol Analysis Module o Deep analysis of email headers including Return-Path, Received headers, Message-ID, Reply-To,DKIM signatures, SPF alignment, and DMARC status. o Detection of anomalies in mail routing, forged sender fields, relay manipulation, and spoofed transmission records. o Validation of whether the email was sent through authorized infrastructure or suspicious relay paths. • Origin Traceability and Location Analysis o Extraction of originating IP addresses from header chains and identification of the earliest reliable sending node. o IP geolocation mapping to estimate the likely country, region, city, ISP, hosting provider, or proxy service associated with the email source. o Correlation with VPN, TOR, open relay, botnet, or cloud-hosted infrastructure indicators where applicable. o Domain intelligence analysis using WHOIS data, DNS records, MX records, hosting fingerprints,and registrar details to identify suspicious sender infrastructure. • Identity Correlation and Attribution Support o Correlation of email indicators with known threat intelligence, blacklists, previous incidents,domain clusters, and repeated fraud campaigns. o Graph-based relationship analysis between sender domains, IP addresses, aliases, reply chains, and linked infrastructure. o Confidence-based investigative assessment to assist in revealing probable sender identity, associated infrastructure, or campaign-level attribution patterns. o Support for flagging whether the email likely originated from a compromised account, spoofed domain, anonymized infrastructure, or direct malicious actor environment. • Alerting, Dashboard, and Forensic Reporting o Real-time alerts for high-risk emails before user interaction or administrative approval. o Analyst dashboard showing fraud score, spoofing indicators, sender trace path, geolocation map,and attribution confidence. o Generation of structured forensic reports for institutional action, legal review, cyber incident response, and support to law enforcement agencies. o Searchable case management view for grouping related fraudulent emails into campaigns. • Privacy, Legal, and Compliance Safeguards o Controlled handling of personal data and metadata in accordance with organizational privacy policies. o Logging, evidence preservation, and chain-of-custody support for investigation purposes. oConfigurable retention and masking mechanisms for sensitive communication data. • Expected Outcomes • Early and accurate detection of fraudulent, spoofed, and phishing-based email attacks. • Improved ability to trace suspicious email origin paths and identify probable source infrastructure. • Enhanced fraud investigation capability through geolocation analysis, domain intelligence, and sender attribution support. • Reduced financial loss, reputational damage, and unauthorized disclosure of confidential information caused by email-based fraud. • Better institutional readiness for cyber incident response, forensic investigation, and enforcement coordination.

SIH26125 SOFTWARE Bharat Electronics Limited

Blockchain-Based Secure Platform for Identity,Access Control, and Digital Asset Management

Theme: Blockchain & Cybersecurity **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Blockchain-Based Secure Platform for Identity,Access Control, and Digital Asset Management. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal [sih.gov.in](sih.gov.in).

SIH26136 SOFTWARE Government Of Maharashtra

Startup friendly public procurement mechanism that enables government departments to identify,pilot, procure,and scale innovative solutions from eligible startups

Theme: Blockchain & Cybersecurity **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

• Problem Description Government departments often face operational problems that could benefit from innovative startup solutions, but conventional procurement processes are generally designed for standardised goods and established vendors. Departments may find it difficult to formulate outcome-based problem statements, discover suitable startups, evaluate novel technologies, structure controlled pilots, manage intellectual property and data, measure pilot results, and transition successful pilots into compliant procurement or scale-up.Startups may struggle with prior-turnover or experience requirements, long sales cycles, unclear payment milestones and limited visibility of departmental demand. The challenge is to create a transparent, competitive and legally compliant innovation-procurement pathway. • Expected Solution / Outcome A structured end-to-end mechanism for challenge identification, startup discovery, eligibility screening, expert evaluation, sandbox or pilot design,milestone-based contracting,performance measurement, payment,independent validation and scale-up decisions. The mechanism should provide standard templates for problem statements, evaluation criteria, pilot agreements, data/IP clauses, cybersecurity, risk management and procurement pathways. It may integrate with recognised startup databases and government e-marketplaces. Expected outcomes include faster discovery and testing of innovative solutions, higher quality pilots, reduced departmental risk, timely startup payments, evidence-based procurement decisions and successful scaling across departments or districts.

SIH26141 SOFTWARE Egreen Quanta

Quantum-Inspired Cyber Threat Detection for Digital Signature Security

Theme: Blockchain & Cybersecurity **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Quantum-Inspired Cyber Threat Detection for Digital Signature Security. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal [sih.gov.in](sih.gov.in).

SIH26145 SOFTWARE National Technical Research Organisation (NTRO)

AI-Based Detection of Cyber Threats in Unidirectional IP Traffic

Theme: Blockchain & Cybersecurity **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AI-Based Detection of Cyber Threats in Unidirectional IP Traffic. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal [sih.gov.in](sih.gov.in).

SIH26148 SOFTWARE National Technical Research Organisation (NTRO)

Creation of scripts/functions with new programming language to commence Computer & Network forensic analysis without triggering security solutions

Theme: Blockchain & Cybersecurity **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Creation of scripts/functions with new programming language to commence Computer & Network forensic analysis without triggering security solutions. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26149 SOFTWARE National Technical Research Organisation (NTRO)

Design and Development of an Integrated Secure Data Erasure and Advanced File Recovery Tool for Digital Forensics and Data Sanitization

Theme: Blockchain & Cybersecurity **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

• Background With the rapid growth of digital storage technologies, organizations, government agencies, law enforcement units, enterprises, and individual users face two major challenges: securely destroying sensitive data to prevent unauthorized recovery and recovering deleted digital evidence during forensic investigations. Existing solutions generally focus on either secure data deletion or file recovery and often support limited storage technologies and file systems. This forces investigators and cybersecurity professionals to use multiple tools, increasing complexity, cost, and operational inefficiencies. Therefore, there is a need for a unified platform that integrates secure data sanitization with advanced forensic-grade file recovery and carving capabilities. • Description The proposed solution aims to develop an integrated software platform consisting of three core modules: (1) Secure Drive Eraser, (2) Secure File & (3) Folder Eraser, and Advanced File Carving and Recovery. The Secure Drive Eraser Module should securely sanitize HDDs, SSDs, USB drives, memory cards, and external storage devices while providing verification mechanisms, audit logging, tamper-resistant reporting, and compliance with industry and government data destruction standards. The Secure File and Folder Eraser Module should enable selective secure deletion of files and folders, remove associated metadata and residual traces, support batch operations, verify erasure success, and provide audit reporting across multiple file systems and operating systems. The Advanced File Carving and Recovery Module should recover deleted files from formatted, damaged, or corrupted media using signature-based, structure-based, and intelligent carving techniques. It should support recovery without file system metadata, fragmented file reconstruction, automatic classification of recovered files, confidence scoring, and comprehensive forensic reporting while preserving evidential integrity. • Expected Solution The expected outcome is an integrated software platform that combines secure data sanitization and forensic recovery capabilities within a single environment. The solution should provide (1) secure drive erasure with verification and reporting, (2) secure file and folder deletion with metadata cleansing, (3) advanced file carving and recovery from formatted media, support for multiple storage devices and file systems, automated classification and validation of recovered files, comprehensive audit logs and forensic reports, a user-friendly graphical interface, and compliance with forensic and data sanitization standards. Expected deliverables include an integrated software tool, (1) Secure Drive Eraser Module, (2) Secure File and Folder Eraser Module, (3) Advanced File Carving and Recovery Module, Reporting and Audit Management System, User Interface Dashboard, validation and testing documentation, user manuals, technical documentation, and performance evaluation reports. The solution should improve secure data disposal practices, reduce the risk of unauthorized data recovery, enhance forensic investigation capabilities, increase recovery rates from damaged storage media, reduce investigation time, improve compliance and auditability, and provide a unified platform for secure sanitization and forensic recovery operations.

SIH26150 SOFTWARE National Technical Research Organisation (NTRO)

Development of a Multi-Vendor DVR/NVR Forensic Analysis Tool for Standardized Acquisition, Recovery, and Analysis of Surveillance Evidence.

Theme: Blockchain & Cybersecurity **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Development of a Multi-Vendor DVR/NVR Forensic Analysis Tool for Standardized Acquisition, Recovery, and Analysis of Surveillance Evidence.. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26151 SOFTWARE

National Technical Research Organisation (NTRO)

Dark web threat actor de-anonymization

Theme: Blockchain & Cybersecurity **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Dark web threat actor de-anonymization. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26155 SOFTWARE

National Technical Research Organisation (NTRO)

AI-Driven Multi-Vendor Network Security Compliance Auditor

Theme: Blockchain & Cybersecurity **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AI-Driven Multi-Vendor Network Security Compliance Auditor. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26156 SOFTWARE

National Technical Research Organisation (NTRO)

Universal Log Pre-processing Framework

Theme: Blockchain & Cybersecurity **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

• Background Modern enterprises generate massive volumes of logs from a wide range of sources, including network devices, servers, operating systems, applications, databases, cloud services, containers, endpoint security tools, identity and access management systems, IoT devices, and other hardware and software platforms. These logs are produced in diverse formats such as Syslog, JSON, XML, CSV, CEF, LEEF, proprietary vendor formats, and application-specific schemas. The diversity of log structures creates significant challenges in centralized monitoring, security operations, compliance reporting, incident investigation, and threat analytics. Security teams often spend substantial effort developing source-specific parsers and normalization rules before the data can be effectively utilized by SIEM, data lake, or machine learning platforms. As organizations adopt hybrid, multi-cloud, and AI-driven environments, the need for a universal and extensible log standard that can accommodate both current and future data sources have become increasingly critical. • Detailed Description Design and develop a Universal Log Pre-processing Framework (ULPF) capable of ingesting, parsing, normalizing, and standardizing logs and events generated by any hardware or software system. The framework should support diverse event sources while preserving the original event data for forensic and compliance purposes. It should transform heterogeneous logs into a unified schema that enables consistent analytics, correlation, visualization, threat hunting, anomaly detection, and machine learning applications. The framework must be scalable, extensible, vendor-agnostic, and suitable for deployment in Big Data environments handling billions of events per day. • Expected Solutions This solution should cover universal event schema and processing framework that enables: a) Preserve complete raw event data without information loss. b) Extract and parse source-specific attributes. c) Normalize fields into a common event taxonomy. d) Maintain traceability between normalized and original events. e) Plug-and-play on boarding of new log sources. f) Unified visibility across enterprise environments. g) Efficient SIEM and Data Lake integration. h) AI/ML-ready security and operational analytics. i) Reduced parser development effort. j) The solution shall be deployable in an air-gapped network. k) Solution may be packaged in a container for making it platform independent. • Current Scope Build a framework that converts any perimeter network device-generated log or eventâ€”regardless of source, format, vendor, or technology into a standardized, lossless, analytics-ready representation for next-generation SIEM and cybersecurity platforms. • Expected Solution/Deliverables for Evaluation • Source Code Link (GitHub/Drive Link) • Readme with Setup Instructions • Architecture Document (Max 2 Pages) • Demo Video (Max 2 Minutes) • Technical Presentation (Max 5 Slides)

SIH26159 SOFTWARE   National Technical Research Organisation (NTRO)

SecureMailScope: AI-Assisted Cryptographic Security Posture Assessment for Secure Email Communications

Theme: Blockchain & Cybersecurity **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: SecureMailScope: AI-Assisted Cryptographic Security Posture Assessment for Secure Email Communications. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26160 SOFTWARE   National Technical Research Organisation (NTRO)

AI-Powered IPsec VPN Protocol Analyzer and Security Assessment Framework

Theme: Blockchain & Cybersecurity **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AI-Powered IPsec VPN Protocol Analyzer and Security Assessment Framework. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26164 SOFTWARE   National Technical Research Organisation (NTRO)

Enterprise Cryptographic Discovery & Analysis Tool (ECDAT)

Theme: Blockchain & Cybersecurity **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Enterprise Cryptographic Discovery & Analysis Tool (ECDAT). Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26182 SOFTWARE   Ministry of Home Affairs

Automated Attribution of Unknown Cryptocurrency Wallets to Nearest Virtual Asset Service Providers (VASPs) through Blockchain Intelligence APIs

Theme: Blockchain & Cybersecurity **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Automated Attribution of Unknown Cryptocurrency Wallets to Nearest Virtual Asset Service Providers (VASPs) through Blockchain Intelligence APIs. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26183 SOFTWARE   Ministry of Home Affairs

Real-Time Identification of Fraud-Linked Cryptocurrency Exchanges from Victim-Reported Suspect Wallet Addresses through Automated Blockchain Analytics

Theme: Blockchain & Cybersecurity **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Real-Time Identification of Fraud-Linked Cryptocurrency Exchanges from Victim-Reported Suspect Wallet Addresses through Automated Blockchain Analytics. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26184 SOFTWARE Ministry of Home Affairs

Development of a Predictive Analytics Framework for Cybercrime Complaints to Forecast Likely Cash Withdrawal Locations in Advance, Enabling Generation of Actionable Intelligence for Timely and Proactive Cybercrime Intervention.

Theme: Blockchain & Cybersecurity **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

• Background The National Cybercrime Reporting Portal is the centralized Portal, which is serving the whole country. Currently, the Portal facilitates citizens in filing complaints, LEAs act on complaints, Banking/Financial Institutions for their actions along with reports/graphs being pulled on daily basis. Presently, the Portal is receiving approximately 8000 complaints on daily basis. The number of complaints has increased manifold during the past months, and this will continue to rise in future. To address the issue of increasing cybercrimes, the proactive approach shall be adopted. • Description This framework focuses on the mitigation of cybercrimes by adopting a proactive approach. The framework's output will enable the prediction of likely cash withdrawal locations, which, in turn, will allow law enforcement agencies (LEAs) at the state and local levels, coordinated by I4C, to implement proactive interventions. These interventions could include deploying special teams or alerting local banks and ATMs in high-risk areas. The intelligence generated would also help banks and financial institutions (FIs) through the Citizen Financial Cyber Fraud Reporting and Management System, enabling faster fund blocking and increasing the chances of recovery. By supporting real-time actionable intelligence sharing across jurisdictions, law enforcement agencies and Banks/FIs will be able to respond faster and more effectively to cyber threats. This approach goes beyond merely reacting to complaints and creates a powerful, data-driven defense against financial cyber frauds, strengthening India's overall cybersecurity posture. Enhancing coordination between law enforcement and financial entities will ensure better detection and prevention of financial crimes, creating a more unified and efficient approach to combating cybercrime. • Key Deliverables Component:- Description a. Predictive Analytics Engine :-AI/ML-based system to analyse historical cybercrime and financial data to predict potential withdrawal hotspots. Features include pattern detection, geospatial risk modelling, and real-time alerts. b. Risk Heatmap Dashboard:-GIS-enabled dashboard visualizing real-time and potential risk zones with drill-down filters by time, location, and crime category etc. c. Law Enforcement Interface:-Secure interface for investigators to access alerts, intelligence reports, and evidence documentation. d. Alert & Notification System:- Real-time notifications to law enforcements, banks, and I4C officers via SMS,email, API, or dashboard triggers.

SIH26189 SOFTWARE Ministry of Home Affairs

AI-Powered Criminal Network Analysis System

Theme: Blockchain & Cybersecurity **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AI-Powered Criminal Network Analysis System. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

🎯 THEME: DISASTER MANAGEMENT

TOTAL: 25 SW: 20 HW: 5

SIH26013 SOFTWARE Ministry of Rural Development

Automated Integration and Intelligent Harmonization of Multi-source Geospatial Data for urban Land Record Management.

Theme: Disaster Management **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Automated Integration and Intelligent Harmonization of Multi-source Geospatial Data for urban Land Record Management.. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26015 SOFTWARE Ministry of Rural Development

Application of Geospatial Techniques for visualization and analysis to interpret Geo-Coded Images to enhance watershed Development Outcomes.

Theme: Disaster Management **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Application of Geospatial Techniques for visualization and analysis to interpret Geo-Coded Images to enhance watershed Development Outcomes.. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26025 HARDWARE Ministry of Coal

Development of an AI-enabled Low Cost Real Time Mine Subsidence Monitoring, Prediction and Early Warning System for Underground Coal Mines in India

Theme: Disaster Management **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Development of an AI-enabled Low Cost Real Time Mine Subsidence Monitoring, Prediction and Early Warning System for Underground Coal Mines in India. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26028 SOFTWARE Ministry of Railways

Dynamic Forecast of Expected Time of Arrival (ETA) for Coaching Trains

Theme: Disaster Management **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Dynamic Forecast of Expected Time of Arrival (ETA) for Coaching Trains. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26029 HARDWARE Ministry of Consumer Affairs, Food & Public Distribution

Automated High-Current Short-Circuit Test System for IEC 60898-1:2015 MCB Compliance.

Theme: Disaster Management **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Automated High-Current Short-Circuit Test System for IEC 60898-1:2015 MCB Compliance.. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26060 SOFTWARE Ministry of Earth Sciences (MoES)

Digital Platform for efficient remote management of Indian Antarctic Research Stations

Theme: Disaster Management **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Digital Platform for efficient remote management of Indian Antarctic Research Stations. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26069 SOFTWARE   Ministry of Earth Sciences (MoES)

National Weather Big Data Analytics Platform

Theme: Disaster Management **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: National Weather Big Data Analytics Platform. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26071 SOFTWARE   Ministry of Earth Sciences (MoES)

AI/ML-Based Integrated heavy rainfall Early Warning and Inundation Prediction System using Satellite, Radar, observational Weather and numerical weather prediction model data.

Theme: Disaster Management **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AI/ML-Based Integrated heavy rainfall Early Warning and Inundation Prediction System using Satellite, Radar, observational Weather and numerical weather prediction model data.. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26072 SOFTWARE   Ministry of Earth Sciences (MoES)

AIML based Nowcasting of thunderstorm and lightning using atmospheric observation including multiple radars, satellite, lightning and model data.

Theme: Disaster Management **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AIML based Nowcasting of thunderstorm and lightning using atmospheric observation including multiple radars, satellite, lightning and model data.. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26073 SOFTWARE Ministry of Earth Sciences (MoES)

AI/ML-Based Intelligent Anomaly Detection for Automatic Weather Stations (AWS)

Theme: Disaster Management **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

• Title SkyGuard AI: Intelligent Real-Time Anomaly Detection System for Temperature, Pressure, and Humidity Sensors in Automatic Weather Stations • Background Automatic Weather Stations (AWS) are critical components of modern meteorological observation networks. These stations continuously monitor atmospheric parameters and provide real-time data for weather forecasting, climate monitoring, disaster management, aviation, agriculture, and scientific research.However, AWS observations often contain anomalies caused by sensor malfunction,communication failures, calibration drift, power fluctuations, harsh environmental conditions, and data corruption.Erroneous observations can significantly impact weather forecasting accuracy and decision-making systems. Traditional threshold-based quality control methods are often insufficient for identifying complex or hidden anomalies in meteorological data streams. • Problem Statement Develop an AI/ML-based intelligent anomaly detection system capable of automatically identifying abnormal, inconsistent, or faulty observations from Automatic Weather Stations in real time using only the following parameters: • Temperature (Â°C) • Atmospheric Pressure (hPa) • Relative Humidity (%) The system should distinguish between genuine meteorological events and sensor/data anomalies while minimizing false alarms and enabling scalable deployment across large weather observation networks. • Objectives • Detect anomalies in real-time AWS data streams. • Identify sensor faults, spikes, frozen values, and communication errors. • Learn normal temporal and seasonal patterns of temperature, pressure, and humidity. • Perform multivariate consistency analysis among atmospheric parameters. • Provide confidence scores and explainable AI-based reasoning for detected anomalies. • Predict possible sensor degradation and maintenance requirements. • Optionally suggest corrected/imputed values for anomalous observations. • Expected Inputs Participants may use historical AWS datasets, simulated anomalies, or streaming sensor data containing the following meteorological parameters: Parameter- Unit Temperature - Â°C Atmospheric Pressure - hPa Relative Humidity - % • Expected Outputs • Real-time anomaly alerts • Severity and confidence scores • Root-cause classification • Visualization dashboard • Sensor health status • Corrected data estimation (optional) • Suggested Technologies • Explainable AI (SHAP/LIME) (Preferable) • Edge AI for low-power deployment on ESP32 • Evaluation Criteria (To be evaluated in anomaly injected data) Criteria - Weightage Innovation & Novelty - 25% Detection Accuracy - 20% Real-Time Capability - 15% Explainability - 10% Scalability - 10% Practical Deployability - 10% Visualization/UI - 5% Energy Efficiency - 5% • Example Use Case An AWS suddenly reports a temperature of 55Â°C with extremely high humidity and abnormal pressure variation while neighboring stations show normal conditions. The AI system should analyze temporal and spatial consistency, identify the reading as a probable sensor anomaly, generate an alert, and suggest corrective action. • Grand Challenge Can AI build a self-aware and self-healing weather observation network capable of delivering trustworthy atmospheric data under all environmental conditions? • Output: Fully executable code with example usage and a document explaining various use cases

SIH26074 SOFTWARE Ministry of Earth Sciences (MoES)

Downscaling of weather forecast from Block level to Panchayat level: Inferring high-resolution plots/ data/ information from low-resolution plot /data /information /variables for agro-meteorological advisory services.

Theme: Disaster Management **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Downscaling of weather forecast from Block level to Panchayat level: Inferring high-resolution plots/ data/ information from low-resolution plot /data /information /variables for agro-meteorological advisory services.. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26078 SOFTWARE Ministry of Earth Sciences (MoES)

AI-Driven Spatio-Temporal Tracking of Extreme Weather Anomalies in Medium-Range Forecasts

Theme: Disaster Management **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AI-Driven Spatio-Temporal Tracking of Extreme Weather Anomalies in Medium-Range Forecasts. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26079 SOFTWARE Ministry of Earth Sciences (MoES)

AI-Based Forecast Bust Detection for Medium-Range Weather Forecasts

Theme: Disaster Management **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AI-Based Forecast Bust Detection for Medium-Range Weather Forecasts. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26080 SOFTWARE Ministry of Earth Sciences (MoES)

Regime-Aware AI Post-Processing of Monsoon Rainfall Forecasts

Theme: Disaster Management **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Regime-Aware AI Post-Processing of Monsoon Rainfall Forecasts. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26082 SOFTWARE Ministry of Earth Sciences (MoES)

Air PollutionWeather Coupled Forecasting System (Delhi NCR Focus)

Theme: Disaster Management **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Air PollutionWeather Coupled Forecasting System (Delhi NCR Focus). Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26083 SOFTWARE Ministry of Earth Sciences (MoES)

Extreme Heatwave Early Warning and Human Thermal Stress Index

Theme: Disaster Management **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

In recent years, the frequency, duration, and intensity of heatwaves across India have escalated sharply due to climate change. However, traditional meteorological warnings rely almost exclusively on ambient dry-bulb temperature thresholds. This creates a critical vulnerability:standard forecasts ignore the deadly compounding effects of relative humidity, wind speed, and solar radiation on the human body. A temperature of 40Â°C at 20% humidity feels vastly different from 40Â°C at 70% humidityâ€”the latter can be fatal. Current public health infrastructure lacks localized, impact-based forecasting that translates raw weather data into actual physiological risk, human thermal stress levels, and projected mortality rates. The challenge is to design an intelligent, localized early warning system that shifts heatwave forecasting from 'what the weather will be' to 'what the weather will do' to human health. Participants need to build a predictive platform that computes a comprehensive Human Thermal Stress Index (integrating temperature, humidity, wind, and radiation) and links it directly to an automated Mortality Risk Index. The system should offer high-resolution forecasts to help municipal corporations, healthcare systems, and disaster management authorities deploy targeted,preemptive interventions. Develop algorithms to calculate advanced heat stress metrics such as the Wet-Bulb Globe Temperature (WBGT), Universal Thermal Climate Index (UTCI), or Heat Index (HI) rather than relying on temperature alone. Integrate historical public health, demographic (e.g., elderly or outdoor worker density), and localized weather data to predict heat-induced mortality and hospitalization spikes 3 to 5 days in advance. A dynamic GIS-mapped dashboard providing colorcoded, hyper-local alerts (Zone/Ward level) paired with actionable, automated public health advisories. An API capable of pushing automated SMS/WhatsApp regional alerts or localized triggers for city administration to initiate heat action plans (e.g., opening cooling centers, adjusting power grids, shifting outdoor work hours).

SIH26084 SOFTWARE Ministry of Earth Sciences (MoES)

Convective scale nowcasting for Thunderstorms, Hail & Cloudbursts (06 hr)

Theme: Disaster Management **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Convective scale nowcasting for Thunderstorms, Hail & Cloudbursts (06 hr). Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26085 SOFTWARE Ministry of Earth Sciences (MoES)

Urban Flood Nowcasting System (Drainage and Rainfall Coupling)

Theme: Disaster Management **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Urban Flood Nowcasting System (Drainage and Rainfall Coupling). Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26161 SOFTWARE National Technical Research Organisation (NTRO)

Dam Break Inundation Modelling Using Hydrodynamic Modelling of any River

Theme: Disaster Management **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Dam Break Inundation Modelling Using Hydrodynamic Modelling of any River. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26177 HARDWARE Qualcomm Inc

A deployable AI-powered autonomous drone that aids search-and-rescue operations by detecting people and hazards, thereby improving responder safety and reducing victim discovery time.

Theme: Disaster Management **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Background India is highly vulnerable to natural disasters including floods, cyclones,earthquakes, landslides, and flash floods, which often result in damaged infrastructure, inaccessible terrain, and delayed rescue operations. During the first few critical hours after a disaster, responders need rapid situational awareness to locate survivors, assess hazards, and prioritize rescue efforts.Traditional ground-based assessments can be slow, dangerous, and resource-intensive, particularly in remote or heavily damaged areas. Autonomous drones equipped with on-device AI can provide real-time aerial intelligence while operating in environments with limited connectivity. Description Develop an autonomous drone system capable of navigating disaster-affected areas and performing real-time detection of survivors and hazards using on-device AI. The drone should use RGB and thermal cameras to identify stranded individuals, detect signs of human presence, and recognize environmental hazards such as fire, floodwaters, damaged structures, exposed electrical lines,debris, landslides, or chemical leaks. The solution must process data locally on the drone to ensure low latency and continued operation even when network connectivity is unavailable. The drone should autonomously map affected regions,generate situational reports, and transmit actionable insights to emergency response teams. This concept aligns with existing edge-AI drone approaches for incident response and disaster assessment. Expected Solution The proposed solution should include some or all of the following: • Autonomous Navigation: GPS-enabled and GPS-denied navigation capabilities using AI, SLAM, and obstacle avoidance for operation in damaged environments. • On-Device AI Inference: Real-time detection of people, survivors, and disaster-related hazards without dependence on cloud connectivity. • Multi-Sensor Fusion: Integration of RGB cameras, thermal cameras, IMU,and GPS sensors for accurate identification and localization of victims. • Hazard Classification: Detection and classification of floods, fires, smoke,debris, unstable structures, landslide zones, and other safety threats. • Geo-Tagged Mapping: Creation of live disaster maps highlighting survivor locations, hazard zones, and safe access routes for rescue teams. • Emergency Alerting: Automatic generation of alerts and prioritized rescue recommendations based on detected risks. • Offline Resilience: Ability to function in communication-constrained environments with optional 5G/Wi-Fi connectivity when available. • Command Center Dashboard: Visualization of drone feeds, detected survivors, hazard markers, and mission status to support disaster management agencies.

SIH26180 HARDWARE Qualcomm Inc

A field-deployable AI-powered Smart Farming Assistant that helps farmers detect crop diseases, pests, nutrient deficiencies, and irrigation needs at an early stage, while improving resilience against droughts, floods, heat waves, and other agricultural risks common in India. The solution should enable higher yields, lower input costs, more efficient water usage, and faster response to emerging threats through real-time on-device intelligence.

Theme: Disaster Management **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Background Agriculture remains a primary livelihood for millions of people in India, but farmers face recurring challenges from droughts, erratic rainfall, floods, pest infestations, crop diseases, heat stress, and soil degradation. Climate variability is increasing the frequency of these risks, affecting crop productivity and farm incomes. Many small and marginal farmers lack access to timely diagnostics and expert advice, particularly in regions with limited internet connectivity. Environmental monitoring, edge AI, and local sensing technologies can help deliver real-time insights directly at the farm level without relying on continuous cloud access.Early detection of crop stress, pest outbreaks, irrigation issues, and adverse environmental conditions can significantly reduce crop losses and improve resilience against agricultural disasters such as droughts, floods, and disease outbreaks. Description Develop a Smart Farming Assistant, an edge AI-powered solution that continuously monitors crop health and environmental conditions directly in the field. Using a combination of cameras, environmental sensors, and on-device AI,the system should identify crop diseases, pest infestations, nutrient deficiencies,water stress, and irrigation requirements in real time.The solution should operate locally on edge devices deployed in farms, enabling rapid analysis and recommendations even in areas with poor connectivity. The system should help farmers make informed decisions about irrigation, pesticide application, fertilizer usage, and crop protection while minimizing water consumption and input costs.The platform should provide actionable alerts and recommendations through a simple mobile or field display interface, enabling farmers to respond quickly to emerging risks before they become large-scale crop failures. Edge AI enables realtime decision-making while reducing dependence on cloud infrastructure. Expected Solution The proposed solution should implement some or all of the following: 1. Crop Health Monitoring • Detect visible signs of crop diseases from leaf and plant images. • Identify nutrient deficiencies through color, texture, and growth analysis. • Monitor crop growth stages and overall field health. 2. Pest Detection and Early Warning • Detect common insect pests and infestation patterns using camera-based AI. • Generate early alerts before infestations spread across fields. • Support targeted intervention rather than blanket pesticide application. 3. Smart Irrigation Management • Monitor soil moisture, temperature, humidity, and weather conditions. • Detect water stress and over-irrigation scenarios. • Recommend optimal irrigation schedules to conserve water. 4. Environmental Risk Monitoring • Track conditions associated with drought, excessive rainfall, flooding, heat stress, and disease outbreaks. • Identify abnormal environmental patterns affecting crop productivity. • Provide localized field-level alerts. 5. Edge AI Processing • Perform image analysis and sensor data processing directly on the device. • Operate in remote areas with limited or intermittent connectivity. • Deliver low-latency recommendations and alerts. 6. Farmer Advisory System • Provide simple recommendations such as: o Irrigate now / delay irrigation o Possible disease detected o Pest activity increasing o Heat-stress warning o Flood-risk alert • Deliver advice through a mobile app, local display, or SMS notifications. 7. Farm Analytics Dashboard • Historical trends in crop health and environmental conditions. • Field-level performance monitoring. • Yield-risk forecasting and decision-support insights. 8. Scalable Deployment • Suitable for smallholder farms, cooperatives, and large agricultural enterprises. • Support integration with weather data, farm equipment, and irrigation systems.

SIH26191 SOFTWARE Ministry of Home Affairs

Intelligent Identification of Hazard-Based Red Zones, Carrying Capacity Assessment, and Immediate Relocation Needs for Vulnerable Habitations

Theme: Disaster Management **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

• Background Indiaâ€™s disaster-prone regions face recurring hazards such as landslides,floods, coastal erosion, and cloudbursts. Vulnerable habitations often remain in unsafe zones, leading to repeated loss of lives and property.Current relocation efforts are largely reactive, initiated after disasters strike, rather than proactively planned. • Description The initiative seeks to develop an intelligent, GIS-enabled decision support platform. This platform will dynamically identify and update multi-hazard Red Zones (areas unsuitable for permanent habitation),assess the carrying capacity of safer alternative sites, and prioritize vulnerable habitations for relocation. The system will integrate hazard intensity, population vulnerability, and disaster history to guide evidence-based decisions. • Expected Solution A robust, AI-driven GIS platform that Maps and updates hazard-based Red Zones in real time, assesses suitability and carrying capacity of safer relocation sites, prioritizes vulnerable habitations for immediate,short-term, and medium-term relocation and provides actionable insights to State Disaster Management Authorities for proactive planning.

SIH26192 SOFTWARE Ministry of Home Affairs

Flash Flood Prediction System for Hilly Regions using Multi-Source Data Theme

Theme: Disaster Management **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Flash Flood Prediction System for Hilly Regions using Multi-Source Data Theme. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26206 SOFTWARE AICTE

Student Innovation

Theme: Disaster Management **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Disaster management includes ideas related to risk mitigation, Planning and management before, after or during a disaster.

SIH26223 HARDWARE AICTE

Student Innovation

Theme: Disaster Management **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Disaster management includes ideas related to risk mitigation, Planning and management before, after or during a disaster.

🎯 THEME: SPACE TECHNOLOGY

TOTAL: 23 SW: 19 HW: 4

SIH26001 SOFTWARE

Ministry of Development of North Eastern Region (MDoNER)

AI-Based early warning and landslide Risk Monitoring System in NER

Theme: Space Technology **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

BACKGROUND:

The North Eastern Region (NER) frequently faces landslides, flash floods, road blockages, and slope failures due to heavy rainfall, fragile terrain, and unplanned hill cutting. These incidents often disrupt connectivity, damage infrastructure, delay emergency response, and isolate remote villages for days. Currently, monitoring of vulnerable zones is mostly reactive and dependent on manual reporting. There is limited use of real-time predictive systems for identifying high-risk zones and issuing early warnings to authorities and local communities. With increasing climate vulnerability in the region, there is a need for an AI-enabled real-time monitoring and prediction system that can help authorities take preventive action before disasters occur.

DESCRIPTION:

This problem statement proposes the development of an AI-powered early warning and monitoring platform capable of predicting and tracking landslide-prone areas in real time across the North Eastern Region. The solution should:

- **a.** Collect and analyse data from: Rainfall patterns Soil moisture sensors Satellite imagery Terrain/slope data Historical landslide records b. Use AI/ML models to identify high-risk zones and predict possible landslide events.
- **c.** Provide real-time alerts to district administrations, disaster management authorities, and local communities.
- **d.** Integrate GIS mapping for visualization of vulnerable roads, villages, and infrastructure.
- **e.** Allow citizens/field officials to upload geo-tagged photos/videos of cracks, slope movement or blocked roads.
- **f.** Generate dashboards showing: • Risk severity levels • Road connectivity status • Weather-linked risk forecasts • Emergency response prioritisation. Support multilingual notifications and low-network/offline functionality for remote areas.

EXPECTED SOLUTION:

A scalable AI-based software platform with: • Real-time GIS dashboard and risk heatmaps • AI/ML-based predictive analytics engine • Mobile/web application for field reporting and alerts. • Integration with IMD weather APIs, satellite feeds, and sensor data • Automated SMS/app-based early warning system • Cloud-based architecture with offline sync support for remote regions The solution should improve disaster preparedness, reduce loss of life and infrastructure damage, and strengthen climate-resilient governance in the North Eastern Region.

SIH26003 SOFTWARE

Ministry of Development of North Eastern Region (MDoNER)

AI-Based Cognitive Gaming and Memory Assistance Platform for Elderly Dementia Patients in North Eastern Region (NER)

Theme: Space Technology **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AI-Based Cognitive Gaming and Memory Assistance Platform for Elderly Dementia Patients in North Eastern Region (NER). Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26004 HARDWARE

Ministry of Development of North Eastern Region (MDoNER)

AI-Assisted Early Detection System for Osteoarthritis (OA) Risk Markers in North Eastern Region (NER)

Theme: Space Technology **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AI-Assisted Early Detection System for Osteoarthritis (OA) Risk Markers in North Eastern Region (NER). Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26009 SOFTWARE

Ministry of Steel

Using AI/ML and Space Technology to Identify Manganese Reserves and Overcome Production Shortfalls.

Theme: Space Technology **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Using AI/ML and Space Technology to Identify Manganese Reserves and Overcome Production Shortfalls.. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26010 HARDWARE

Ministry of Rural Development

Survey/Resurvey of Rural Agricultural Land in India

Theme: Space Technology **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

BACKGROUND:
Historically, land surveys in rural India were conducted using conventional chain and tape methods, many of which date back several decades or even the colonial period. Over time, multiple issues emerged such as Boundary changes due to inheritance and informal partition, Unrecorded land transactions, Encroachments and overlapping claims, Errors in cadastral maps, Mismatch between textual records and spatial maps, Absence of updated mutation records and inconsistent land classifications. These deficiencies have resulted in prolonged legal disputes, reduced agricultural productivity, and administrative inefficiencies. Land-related disputes reportedly account for a major share of civil litigation in India.

DETAILED DESCRIPTION:
A comprehensive survey/resurvey program is necessary to establish accurate land ownership, Update cadastral maps, Reduce land disputes,Enable transparent land governance, Support precision agriculture, Improve rural planning, Facilitate digital land administration and Ensure effective implementation of government schemes.Modern technologies such as Drone mapping, Differential GPS (DGPS), GIS platforms, Satellite imagery, CORS, Mobile-based field verification can significanfly improve accuracy, speed, and transparency in rural land management.

EXPECTED SOLUTION:
Technology-Driven Land Survey and Resurvey be implemented using modern survey technologies for accurate mapping of agricultural land parcels such as Drone-based aerial surveys, Real-Time Kinematic (RTK) GPS and DGPS systems, GIS-enabled cadastral mapping and Geo-referenced parcel identification. Developing a unified digital land information system integrating Record of Rights(RoR), Mutation records, Registration databases, Survey maps, Ownership history,precise Geo-coordinates of land parcels. This integration should enable real-time updating and verification of land ownership.

SIH26011 SOFTWARE

Ministry of Rural Development

3D ULPIN Generation and vertical Property Mapping SYstem

Theme: Space Technology **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: 3D ULPIN Generation and vertical Property Mapping SYstem. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26046 SOFTWARE

Ministry of Ayush

AIIA Clinical Trials Dashboard - a real-time, cloud-based, GCP-compliant Clinical Trial Management System (CTMS) for Ayurveda research, with CDISC/FHIR-interoperable data, role-based KPIs, and integrated ethics, regulatory (CTRI / NDCT Rules 2019) and pharmacovigilance tracking.

Theme: Space Technology **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AIIA Clinical Trials Dashboard - a real-time, cloud-based, GCP-compliant Clinical Trial Management System (CTMS) for Ayurveda research, with CDISC/FHIR-interoperable data, role-based KPIs, and integrated ethics, regulatory (CTRI / NDCT Rules 2019) and pharmacovigilance tracking.. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26063 SOFTWARE

Ministry of Earth Sciences (MoES)

Integrated Polar Science Outreach, Knowledge Repository and Media Dissemination Portal

Theme: Space Technology **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Integrated Polar Science Outreach, Knowledge Repository and Media Dissemination Portal. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26066 SOFTWARE

Ministry of Earth Sciences (MoES)

OceanEmbed - Satellite Embedding-Based Deep Learning Framework for Reconstruction of Subsurface Ocean Temperature from Surface Satellite Observations.

Theme: Space Technology **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: OceanEmbed - Satellite Embedding-Based Deep Learning Framework for Reconstruction of Subsurface Ocean Temperature from Surface Satellite Observations.. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26068 SOFTWARE Ministry of Earth Sciences (MoES)

WeatherGPT: Conversational AI for Weather Forecasting, Alerts, and Climate Information

Theme: Space Technology **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

• Background Weather information is often distributed through multiple portals, bulletins, satellite products, and forecast systems, making it difficult for common users, researchers, disaster managers, and government agencies to quickly obtain actionable insights. There is a need for an intelligent conversational platform that can provide real-time weather information, forecasts, warnings, climate analysis, and decision support in natural language. • Objective Develop an AI-powered chatbot platform named WeatherGPT that integrates meteorological datasets, forecasting models, and disaster warning systems to provide accurate, contextual, and multilingual weather intelligence through conversational interfaces. • Key Features 1. Real-time weather information retrieval. 2. Natural language querying for weather forecasts. 3. Integration with numerical weather prediction (NWP) models such as GFS/WRF. 4. Extreme weather alerts and early warning dissemination. 5. Location-based forecasting and advisory generation. 6. Multilingual support for Indian languages. 7. Climate trend and historical weather analysis. 8. Voice-enabled interaction for rural accessibility. • Expected Solution Participants should develop: • A mobile-based conversational AI platform. • Backend integration with meteorological databases, website and APIs. • AI/LLM-based query understanding engine. • Scalable architecture supporting real-time data ingestion. • Suggested Technology Stack • Python / FastAPI / Node.js • MQTT / WIS2.0 / WebSocket • LLMs (OpenAI, Llama, Gemini, etc.) • GIS tools and weather APIs • PostgreSQL / MongoDB • Docker / Kubernetes • Expected Outcomes • Faster dissemination of weather information. • Improved public accessibility to forecasts. • Better disaster preparedness and response. • Intelligent weather decision-support system for agriculture, aviation, marine, and urban planning. • Possible Use Cases • Farmers seeking crop-weather advisories. • Aviation weather briefing. • Flood/cyclone warning dissemination. • Smart city weather monitoring. • Climate analytics for researchers. • Evaluation Parameters • Accuracy and relevance. • Response latency. • Multilingual capability. • User interface and accessibility. • Scalability and innovation. • Integration with real-time meteorological systems. • Voice-enabled interaction for rural accessibility

SIH26077 SOFTWARE Ministry of Earth Sciences (MoES)

AI-Driven Hyper-Local Early Warning System for Severe Weather Nowcasting

Theme: Space Technology **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

• Problem Statement India is highly vulnerable to rapidly intensifying, localized extreme weather events such as cloudbursts, severe thunderstorms, and flash floods. Traditional physics-based Numerical Weather Prediction (NWP) models often suffer from computational latency and struggle to capture the rapid, small-scale atmospheric changes that preceded these events. There is a critical need for a real-time, hyper-local early warning system capable of 'nowcasting' severe weather 2 to 6 hours before impact, providing actionable lead time for disaster management. • Proposed Solution We propose an advanced AI predictive engine designed for high-precision severe-weather nowcasting. Specifically, the system simultaneously predicts the onset of highly localized, rapidly intensifying events, namely severe thunderstorms, cloudbursts, and the subsequent flash floods,with an actionable lead time of 2 to 6 hours. Instead of relying on computationally intensive thermodynamic simulations, the system utilizes a spatiotemporal deep learning architecture to recognize the complex, multivariate atmospheric signatures that precede these extreme events. A critical component of this methodology is storm nowcasting using variations in integrated water vapor (IWV). By tracking rapid spatial and temporal accumulations of IWV, the model accurately identifies the concentrated moisture pools required for heavy precipitation. To predict multiple extreme events simultaneously, the engine employs a multi-task learning approach. A shared neural network backbone extracts foundational atmospheric features (moisture, instability, and lift) from the input grids. The network then branches into distinct output layers, allowing a single unified model to generate hyper-local probability risk maps for thunderstorms, cloudbursts, and flash floods simultaneously, entirely bypassing the computational latency typical of traditional numerical weather prediction (NWP) models. Predictive Matrix: Key Atmospheric Variables Severe convective storms require three primary ingredients: moisture, instability, and lift. Our AI model tracks the critical precursors across all three categories to ensure high accuracy and low false-alarm rates: ? Moisture Availability (The Fuel): The cornerstone of our storm nowcasting is the capture of integrated water vapor (IWV) variations. By tracking rapid spatial and temporal accumulations of IWV from satellites, the model identifies the concentrated moisture pools that trigger localized cloudbursts. ? Atmospheric Instability (The Energy): The model assesses the atmosphere's thermal profile to determine if it is buoyant enough to support explosive vertical cloud growth. High Convective Available Potential Energy (CAPE) paired with eroding Convective Inhibition (CIN) serves as a prime indicator of impending severe thunderstorms. Kinematics and Lift (The Trigger & Structure): Low-level convergence (wind vectors colliding at the surface) forces air upward, initiating the development of a storm cell. Furthermore, tracking vertical wind shear (changes in wind speed/direction with altitude) helps the model predict whether a storm will move quickly or remain stationary. Observational Signatures: Rapid cooling of cloud tops, measured as the Cloud Top Temperature(CTT) Drop Rate, provides real-time validation of explosive vertical updrafts within the system. Topographic Dynamics (The Flood Catalyst): To accurately predict flash floods, the AI overlays the atmospheric probability maps onto a high-resolution Digital Elevation Model (DEM). This allows the system to calculate how terrain slope, elevation, and natural drainage basins will channel the extreme precipitation generated by a predicted cloudburst.To capture these predictors with hyper-local accuracy, the model fuses multi-modal, high resolution datasets: IMDAA Reanalysis Data (Historical Baseline & Thermodynamics): Multi-level air temperature, specific humidity profiles (for calculating CAPE/CIN), geopotential height, and U/V wind components (for calculating shear and convergence). Satellite Observations (INSAT-3D/3DR via MOSDAC): Water Vapor (WV) Channels. This is essential for deriving real-time Integrated Water Vapor (IWV) fluctuations necessary for our storm nowcasting. Thermal Infrared (TIR) Channels: Utilized to calculate the rapid Cloud Top Temperature (CTT) drop rate. Quantitative Precipitation Estimation (QPE): Satellite-derived precipitation estimates are used to monitor real-time rainfall intensity, serving as a reliable, openly accessible alternative to ground-based radar. Digital Elevation Model (DEM): High-resolution topographical data (such as ISRO's CartoDEM or SRTM) provides a static baseline of elevation, slope, and surface drainage networks, enabling translation of atmospheric cloudburst predictions into actionable flash flood warnings on the ground. • Technical Methodology ? Data Fusion & Alignment: Raw data from IMDAA reanalysis, INSAT-3D/3DR satellite observations, and high-resolution Digital Elevation Models (DEM) are ingested, normalized, and mapped onto a unified spatiotemporal grid (e.g., using multi-dimensional array structures). This ensures that all dynamic atmospheric predictorsâ€”such as specific humidity and cloud-top temperaturesâ€”and static surface variables align geographically and chronologically for seamless multimodal processing. ? Multi-Variate Feature Extraction & Multi-Task Inference: A shared multi-modal spatiotemporal transformer network continuously analyzes real-time satellite grids, specifically tracking critical IWV variations and CTT drop rates, against the IMDAA-derived thermodynamic baselines using cross-attention mechanisms. Utilizing a Multi-Task Learning (MTL) architecture,the network branches into distinct output 'heads.' This allows the unified model to simultaneously process the aligned data and generate distinct, hyper-local probability maps for severe thunderstorms, cloudbursts, and flash floods without computational bottlenecking. ? Automated Alerting: When the predictive matrix breaches the signature thresholds of a severe event, the engine generates a spatial risk map and pushes automated, categorized alerts via a lightweight API. • Expected Solution The final deliverable for the Smart India Hackathon will be a fully functional, real-time prototypeof the AI-Driven Hyper-Local Early Warning System. At its core is a deployed multi-task inference engine that continuously ingests live INSAT satellite data and IMDAA thermodynamic baselines to simultaneously generate predictive risk maps for severe thunderstorms, cloudbursts, and flash floods within a 2 to 6-hour predictive window. This backend integrates with an interactive, webbased spatial dashboard designed for disaster management authorities, featuring dynamic risk maps overlaid on a Digital Elevation Model (DEM) and an Explainable AI (XAI) module that transparently displays meteorological triggers. Finally, an automated API will translate these predictive insights into immediate, categorized alerts sent directly to first responders and vulnerable communities the moment critical thresholds are breached.

SIH26142 SOFTWARE National Technical Research Organisation (NTRO)

Deep Learning Based Super Resolution Mapping (SRM) from Medium Resolution Satellite Imageries

Theme: Space Technology **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

• Background Medium-resolution satellite imagery, typically ranging from 10 to 30 meters, is widely used in change detection, agriculture, land-cover mapping, disaster monitoring, and urban planning because it offers broad coverage and frequent revisit time. However, the spatial detail is often insufficient for fine-scale analysis, such as identifying small buildings, narrow roads, field boundaries, or localized damage assessment. This creates a need for advanced deep learning based generative enhancement techniques that can extract greater value from existing Earth observation data. • Description Medium-resolution satellite imagery, usually ranging from 10 to 30 meters, is widely used in remote sensing for agriculture monitoring, land-cover mapping, urban planning, disaster assessment, and environmental observation because it provides large-area coverage and frequent revisit capability. However, its spatial resolution is often not sufficient to clearly identify fine details such as narrow roads, small buildings, field boundaries, water edges, or localized damage. This limitation reduces the accuracy and confidence of interpretation and decision-making in applications that require detailed ground-level information. Generative AI super-resolution addresses this problem by using advanced models such as GANs, diffusion models, and deep neural networks to enhance medium-resolution satellite images into sharper and more information-rich finer outputs. These models learn spatial textures, patterns, edges, and spectral relationships from training data containing both medium-resolution and high-resolution image pairs. The goal is not simply to make the image visually clearer, but to reconstruct useful fine-scale details while preserving the original geographic and spectral consistency of the satellite data. The expected solution is a robust AI-based super-resolution framework that can take medium-resolution satellite imagery as input, perform pre-processing, apply a trained generative model, and produce an enhanced spatial resolution image, suitable for analysis. The system should improve feature visibility, support better classification, change detection, crop monitoring, urban mapping, and disaster response. At the same time, it must clearly manage uncertainty because some reconstructed details are inferred by the model and not directly observed. Therefore, validation against high-resolution reference data is essential to ensure that the enhanced outputs are scientifically reliable and useful for real-world remote sensing applications. • Expected Solution The expected solution is a robust super-resolution framework model based on the choice of participating team (Transformers/Generative/CNN etc.) that can transform the input medium-resolution satellite imagery (10m Sentinel-2 Satellite Imagery) into sharper, information-rich products (<4m) while preserving geospatial and spectral consistency. The solution should include pre-processing, model training with paired datasets, accuracy assessment, and validation against high-resolution references. Ideally, it should support applications such as crop monitoring, urban analysis, and disaster assessment. The final outcome should improve in-terms of interpretability and analytical utility, while clearly accounting for uncertainty and error components.

SIH26143 SOFTWARE National Technical Research Organisation (NTRO)

Leveraging satellite imagery to determine Oil spills at sea along with AIS data correlations to identify vessel responsible for the spill.

Theme: Space Technology **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Leveraging satellite imagery to determine Oil spills at sea along with AIS data correlations to identify vessel responsible for the spill.. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal [sih.gov.in](sih.gov.in).

SIH26153 SOFTWARE National Technical Research Organisation (NTRO)

AI based Network Attack Forecasting from Network Traffic Data

Theme: Space Technology **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

• Background This challenge seeks AI systems capable of learning network behaviour, anticipating attacker progression and supporting proactive cyber defence using the emerging concept of World Models. Design and develop a software prototype that learns the evolving state of a computer network from traffic telemetry and predicts the likelihood and progression of malicious activity before compromise is completed. The solution should ingest network traffic, learn temporal behaviour, forecast future attack states and provide interpretable decision support for defenders. Solutions should demonstrate applicability to enterprise environments and Critical Information Infrastructure. • Represent network state using feature vectors or graphs. • Learn state-transition dynamics using sequence models (LSTM, Transformer), Graph Neural Networks, latent state models or other AI techniques. • Forecast future network states and estimate the probability of attacker progression. • Map predicted behaviour to recognised attack stages (e.g. MITRE ATT&CK). • Provide explain ability using attention mechanisms, feature attribution or equivalent techniques • Detailed Description Participants are encouraged to build world models based AI systems that move beyond static intrusion classification towards predictive cyber defence. The solution may utilise flow records, packet captures, authentication logs or other publicly available cybersecurity telemetry. It should model temporal relationships, infer evolving network state, predict future attack progression and present meaningful explanations for its predictions. Traditional machine learning classifiers applied to network traffic treat each flow in isolation and map it to a binary benign/malicious label. This discards the temporal and causal structure of an infiltration: the sequence in which ports are probed, the pattern in which SYN flags precede ACK floods, the inter-arrival timing of reconnaissance packets before lateral movement begins. An infiltration is a process unfolding over time, not a single anomalous packet. • World Models â€" AI architectures that learn an internal causal simulation of how environment states evolve â€" offer a fundamentally different approach. Rather than classifying traffic, a world model learns the transition dynamics P(S_t+1 | S_t): given the current observed network state (active flows, flag distributions, port activity, packet timing), what is the probability distribution over future states. This enables forward simulation: roll out K steps ahead and identify whether the current trajectory converges to an infiltration state, before the attacker completes the kill chain. 1. Input Data â€" Two Levels of Traffic Feature Teams must work with both flow-level and packet-level features drawn from open-source network traffic datasets: • Flow-level features (NetFlow / IPFIX format): source and destination IP/port pairs, TCP flag bitmask (SYN, ACK, FIN, RST, PSH, URG), protocol, bytes transferred per flow, packets per flow, flow duration, inter-arrival time (IAT) statistics (mean, variance, max), and bidirectional flow ratios. • Packet-level features (PCAP-derived): Time-To-Live (TTL) values and their variance across a session, TCP window size, IP fragment flags, payload size distribution, port scan signatures (sequential or randomised port access patterns), and retransmission counts. The combination of both levels is required because flow-level features capture aggregate behaviour (a SYN flood) while packet-level features expose timing and sequencing patterns (a slow reconnaissance scan designed to evade flow-based thresholds). 2. World Model Architecture The core deliverable is a learned model of network state transition dynamics â€" not a static classifier. The model must: • Represent network state as a structured feature vector or graph encoding active flows at time t. • Learn P(S_t+1 | S_t) â€" the probability distribution over the next network state given the current state â€" using a sequence model such as an LSTM, Temporal Transformer, or Graph Neural Network (GNN) operating over time-windowed traffic observations. • Be trained on labelled open-source datasets using supervised dynamics learning, where ground-truth state transitions are derived from the attack timeline annotations in the dataset. • Generalise to unseen attack patterns â€" not merely memorize signatures from the training set. 3. Infiltration Prediction and Attack Stage Mapping The world model must support forward simulation: given current observed traffic, roll out K steps and output • A time-series probability score: likelihood of infiltration in the next K time windows. • Predicted attack stage: mapping to MITRE ATT&CK phases â€" Reconnaissance, Initial Access, Lateral Movement, Command & Control, or Exfiltration â€" based on the predicted future state. • Driving features: which specific flags, ports, or flow patterns are contributing most to the infiltration prediction (via attention weights or SHAP values). The approaches are provided only as examples and are not mandatory. Teams are free to propose alternative architectures that satisfy the objectives. • Expected Solution(Indicative) A software-based, fully open-source solution is expected. The solution may include: • A feature extraction pipeline that ingests CIC-IDS-2018 or CTU-13 CSV flow records and/or raw PCAP files (parsed using Scapy or PyShark) and outputs a timestamped, normalised feature matrix covering both flow-level and packet-level attributes described above. • A trained world model (LSTM, Transformer, or GNN architecture) that demonstrably learns traffic state transition dynamics â€" not a static input-output classifier. Training scripts, model weights, and a reproducible training configuration must be included. • An infiltration prediction engine that performs K-step forward simulation from a current traffic snapshot and outputs: infiltration probability score, predicted MITRE ATT&CK stage, and top contributing traffic features. • An explainability output for each prediction â€" using SHAP values or model attention weights â€" identifying which flags, ports, or flow statistics are driving the prediction. Black-box outputs without interpretability are not acceptable. • A working demonstration interface (Streamlit, Flask web app, or CLI) that accepts a PCAP or CSV file as input, runs the world model inference, and displays the infiltration probability timeline, flagged flows, and attack stage annotations. The interface must run fully offline without cloud API dependencies. • Benchmark results comparing model performance (F1 score, precision, recall, false positive rate) against a logistic regression baseline trained on the same features, demonstrating that the world model's temporal dynamics learning provides measurable improvement. • Expected Solution/Deliverables for Evaluation

• Source Code Link (GitHub/Drive Link) • Readme with Setup Instructions • Architecture Document (Max 2 Pages) • Demo Video (Max 2 Minutes) • Technical Presentation (Max 5 Slides) .

SIH26162 SOFTWARE National Technical Research Organisation (NTRO)

AI-Based Detection and Classification of Industrial Fires and Persistent Thermal Sources Using NASA FIRMS, OSM & Satellite Data

Theme: Space Technology **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AI-Based Detection and Classification of Industrial Fires and Persistent Thermal Sources Using NASA FIRMS, OSM & Satellite Data. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26166 SOFTWARE Indian Space Research Organisation(ISRO)

Multi-modal, Sun angle and scale invariant image correspondence using Chandrayaan-2 optical images (OHRC, TMC and IIRS)

Theme: Space Technology **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Multi-modal, Sun angle and scale invariant image correspondence using Chandrayaan-2 optical images (OHRC, TMC and IIRS). Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26167 SOFTWARE Indian Space Research Organisation(ISRO)

SatQuery AI - An Interactive Vision-Language Assistant for Multimodal Remote Sensing Image Analysis through Text Queries

Theme: Space Technology **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Background Remote-sensing imagery is widely used for agricultural monitoring, disaster management, urban planning, forest monitoring, water-resource assessment, infrastructure mapping, and environmental analysis. However, most existing remote-sensing AI solutions are developed as isolated applications for a single predefined task, such as land-cover classification, object detection, visual question answering, or change detection. These systems often require users to understand satellite-data characteristics, GIS workflows, model selection, and task-specific parameters. Consequently, non-expert users may find it difficult to obtain meaningful information from satellite imagery through simple natural-language queries. Many operational remote-sensing questions cannot always be answered reliably using a single optical image. Relevant information may be distributed across paired or multiple observations acquired at different times or by different sensors. Optical and multispectral imagery provides spectral and contextual information, whereas synthetic aperture radar (SAR) provides complementary structural information and supports day-and-night acquisition through cloud cover. Multitemporal image pairs are required to identify and interpret changes over time, while co-registered opticalâ€“SAR pairs can provide more complete and reliable information than either modality alone. A general-purpose large language model (LLM) or vision-language model (VLM) cannot be expected to perform these specialised tasks reliably without adaptation to remote-sensing imagery, sensor characteristics, and domain-specific terminology. The proposed solution must therefore include remote-sensing fine-tuning or domain adaptation and may employ multiple specialised models for different tasks. BigEarthNet.txt will serve as the primary dataset for adapting imageâ€“text representations to multisensor remote-sensing data. VRSBench and RSVQA will be used to evaluate single-image captioning, grounding, and visual question answering, while CDVQA will be used to evaluate multitemporal change-based visual question answering. The novelty of SatQuery AI lies in its agentic, query-driven framework. Instead of applying a single generic VLM, the system selects and executes suitable remote-sensing specialist models, validates inputs, combines their outputs, and returns an evidence-grounded response. Description The objective is to develop SatQuery AI, a software-based agentic vision-language assistant for analysing single and paired remote-sensing images through natural-language queries. Single-image understanding is a mandatory baseline, while the principal focus is joint reasoning over paired cross-modal and multitemporal imagery. Defined Input Scope • Single image: One optical/multispectral or SAR image for captioning, visual question answering, and text-guided region grounding. • Cross-modal pair: Co-registered optical/multispectral and SAR images of the same geographic area for joint information extraction and cross-modal analysis. • Bi-temporal pair: Two spatially corresponding images of the same geographic area acquired at different times for change detection, change description, and change-based visual question answering. • Supported formats: GeoTIFF or TIFF for geospatial imagery. PNG and JPEG inputs may be accepted only for the prescribed public benchmark datasets. Mandatory Functional Scope • Remote-sensing adaptation: At least one visual or vision-language component must be fine-tuned or otherwise adapted using BigEarthNet.txt or the any open source training data. • Single-image baseline: Visual question answering shall be mandatory. Each solution must additionally implement either captioning/scene description or text-guided region grounding. • Multi-image change analysis: Change description or change-based visual question answering from a bi-temporal image pair shall be mandatory. A spatial change map may also be generated where reference masks are available. • Cross-modal pair analysis: The system must extract complementary information from a co-registered optical/multispectral and SAR image pair. • Agentic orchestration: The system must automatically select, sequence, and execute the appropriate specialist models or tools according to the query and input configuration. Representative Queries • 'Describe the land-cover and major objects visible in this image.' • 'Highlight the water body referred to in the query.' • 'What changed between these two dates, and where did the change occur?' • 'Use the optical and SAR images together to identify built-up and water-covered regions.' • 'Has the built-up area increased, decreased, or remained unchanged?' Agentic Model and Tool Orchestration The system may use multiple specialised components, such as a remote-sensing VQA or captioning model, a grounding model, a change-understanding or change-VQA model, and an opticalâ€“SAR fusion or information-extraction model. • interpret the query and classify the requested task; • check the number, modality, format, metadata, and compatibility of the input images; • select one or more models or tools from a predefined registry; • configure only permitted task parameters and execute the selected workflow; • combine textual and spatial outputs, estimate confidence, and return visual evidence; and • provide an auditable execution summary containing the selected task, model/tool names, and key parameters. The controller may perform internal task planning; however, only the observable execution trace, including the selected task, models or tools, permitted parameters, and outputs will be evaluated. Internal reasoning text is neither required nor evaluated. Expected Solution The expected solution is an interactive GUI or web application with an agentic remote-sensing AI backend. It should accept supported image inputs and natural-language queries, select the appropriate specialist workflow, and return evidence-grounded textual and visual results. The solution should include: • Input upload and compatibility checking. • A remote-sensing-adapted vision-language component. • Specialist tools for VQA, captioning or grounding, change understanding, and opticalâ€“SAR analysis. • An agentic controller for task routing, tool execution, and output integration. • Visual evidence, confidence information, execution summaries, and downloadable reports. Each solution must demonstrate single-image VQA, one additional single-image task, multitemporal change understanding, opticalâ€“SAR paired-image analysis, and agentic model/tool orchestration. A generic LLM or VLM without remote-sensing adaptation will not satisfy the requirements. Deliverables An interactive GUI or web application with an agentic remote-sensing AI backend, Codes and models including test and demonstration. Implementation Scope The system shall support single optical/multispectral or SAR images, co-registered opticalâ€“SAR pairs, and bi-temporal pairs in GeoTIFF/TIFF or approved benchmark formats. It must perform single-image VQA, one additional single-image task, change analysis, opticalâ€“SAR joint analysis, and agentic model/tool selection through an interactive GUI or web application. Evaluation/Judging Criteria Final evaluation will use prescribed public benchmark test subsets and an ISRO/SAC evaluation dataset. Scores will be normalised before combining different metrics. Add 'Evaluation/Judging Criteria' table here Public benchmarks will be evaluated using the prescribed test splits. The ISRO/SAC evaluation set will contain pre-georeferenced and co-registered Cartosat-2S optical and RISAT SAR image pairs, with task-specific reference answers, labels, bounding boxes, or masks, as applicable. Evaluation annotations will not be disclosed to participating teams.

SIH26169 SOFTWARE Indian Space Research Organisation(ISRO)

Development of an AI-Based Virtual Camera Tracking System for Coarse Alignment of Mobile Free Space Optical Communication (FSOC) Terminals

Theme: Space Technology **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Development of an AI-Based Virtual Camera Tracking System for Coarse Alignment of Mobile Free Space Optical Communication (FSOC) Terminals. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26175 SOFTWARE Indian Space Research Organisation(ISRO)

DepthWizard - Single-View Height Estimation and 3D Flythrough

Theme: Space Technology **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Background Accurate Digital Elevation Models (DEMs) and Digital Surface Models (DSMs) are fundamental to urban planning, disaster management, and military reconnaissance. Traditionally, elevation data is acquired through stereo-imaging pairs, LiDAR, or Interferometric Synthetic Aperture Radar (InSAR). These approaches can be cost-prohibitive, dependent on specific sensor availability, and computationally intensive. Single-view height estimation offers an agile alternative, but foundational monocular depth models are trained largely on natural egocentric imagery and predict relative depth. When applied to remote sensing, they face domain gaps, structural variations, and a lack of absolute-scale mapping. Converting relative depth into metric elevation remains a critical challenge, alongside the operational need to transform static elevation profiles into interactive 3D assets that can be navigated in real time. Description Develop an end-to-end software pipeline that transforms single-view optical RGB remote-sensing images into high-precision elevation maps. The framework must support both non-georeferenced and georeferenced imagery. • Non-Georeferenced RGB Imagery (for example, PNG or JPG): Produce a Relative Digital Surface Model (rDSM) for images without spatial metadata. • Georeferenced RGB Imagery (for example, GeoTIFF): Produce an Absolute Digital Surface Model (DSM) with metric height values for images containing coordinate-system metadata. The solution should use a pre-trained monocular depth-estimation backbone to generate initial relative-depth maps. For georeferenced imagery, a lower-resolution DEM source such as SRTM or a limited set of Ground Control Points may be used to map scale-agnostic depth features to absolute metric elevations. For non-georeferenced imagery, relative height may be used directly in the visualization stage. After computing the elevation map, the system should project the original optical image onto a generated 3D terrain mesh and integrate the result with a rendering engine such as Unity, Three.js, or Babylon.js. The interface should support seamless first-person navigation and analysis of structural heights and slopes from arbitrary aerial perspectives. Key Milestones • Elevation Extraction: Use a robust pre-trained monocular depth model to extract geometric and structural representations from single-view optical imagery. • Scale Calibration: Develop a module that converts relative depth to absolute height using scene-level statistics, low-resolution DEMs, semantic priors, or minimal Ground Control Points for georeferenced inputs. • Visualization Layer: Build an immersive, preferably interactive, rendering pipeline that converts the optical texture and derived depth map into a navigable 3D environment deployable as a standalone application. Evaluation Criteria: • DSM Estimation - Accuracy and Validation (50%): Evaluate RMSE, MAE, and correlation against LiDAR or reference data, including performance stability across urban, sparse, hilly, and forested landscapes. • Visualization - Rendering Quality and User Experience (50%): Assess projection accuracy, visual fidelity, navigability of the 3D flythrough, interface intuitiveness, software stability, and successful standalone deployment. Expected Solution Deliver a fully integrated software suite with complete source code and technical documentation. The solution must be deployable as a unified module containing the following components: • Elevation Estimation Module: Accept single-view optical satellite imagery in PNG, JPG, or TIFF format and output a high-fidelity DSM in a standard geospatial format. • Interactive Visualization Platform: Provide a user-friendly 3D flythrough experience that lets users upload imagery, visualize reconstructed terrain, and validate estimated height values against reference datasets.

SIH26176 SOFTWARE Indian Space Research Organisation(ISRO)

ORCA Marine EcOsystem Reasoning with Collaborative Agents

Theme: Space Technology **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Background The marine ecosystem plays a vital role in supporting livelihoods, food security, biodiversity, maritime transportation, coastal resilience, and the blue economy. Every day, vast volumes of satellite Earth Observation and oceanographic data, including Sea Surface Temperature (SST), chlorophyll concentration, and weather forecasts, are generated by ISRO and other global agencies. Marine stakeholders such as fishermen, researchers, coastal authorities, disaster management agencies, and maritime operators rely on timely access to oceanographic and meteorological information for operational planning and decision-making. As the volume, diversity, and complexity of marine data continue to increase, there is a growing need for an intelligent conversational platform that enables users to interact naturally with marine information, ask questions, explore scenarios, and receive synthesized, evidence-based recommendations tailored to their context. Advances in Agentic AI, conversational intelligence, and geospatial technologies present an opportunity to fundamentally transform how marine information is accessed and utilized. By integrating satellite Earth Observation data with autonomous AI agents, intelligent conversational decision-support systems should be able to answer questions such as 'Where is the nearest Potential Fishing Zone today?', 'Is it safe to venture into the sea tomorrow morning?', or 'What are the tide, weather, and sea conditions near my fishing location?' and provide explainable, context-aware recommendations. Description Develop an Agentic AI-powered conversational platform that enables users to access, analyze, and reason over marine information using natural language. The platform should autonomously interpret user intent, decompose complex requests into executable tasks, coordinate multiple specialized AI agents, retrieve relevant marine and geospatial datasets, perform spatial-temporal reasoning, and synthesize actionable recommendations through a conversational interface. The solution should be capable of integrating information from multiple sources, including satellite Earth Observation products, GIS layers, weather services, oceanographic observations, and marine advisories available in the public domain. Typical user queries include: • Where is the nearest Potential Fishing Zone (PFZ) today? • Is it safe to venture into the sea tomorrow morning? • What are the tide, weather, and sea conditions near my fishing location? • Are there any lightning or cyclone alerts in my area? • Which regions show high chlorophyll concentration and favourable sea surface temperature? • What is the safest route for a fishing vessel considering weather and sea-state conditions? • Why has fish productivity declined in a particular coastal region? • Which fishing zones should be avoided due to hazardous marine conditions or geofencing restrictions? The platform should not merely retrieve information from individual datasets but intelligently correlate observations from multiple sources, explain the reasoning behind its recommendations, and present insights through conversational responses, maps, alerts, and interactive geospatial visualizations. Expected Solution Participants are expected to develop an Agentic AI-powered Marine Intelligence Platform that leverages collaborative AI agents, geospatial technologies, and satellite Earth Observation data to provide intelligent conversational decision support. The solution should demonstrate the core principles of Agentic AI, including autonomous planning, reasoning, tool selection, task execution, collaboration among specialized agents, and explainable decision-making. The platform should be capable of: • Understanding user intent expressed in natural language. • Automatically identifying the language of the user's query and responding in the same language, with emphasis on supporting Indian regional languages. • Supporting contextual, multi-turn conversations that enable users to refine queries and explore related scenarios. • Autonomously discovering, retrieving, and integrating relevant satellite, marine, meteorological, and geospatial datasets. • Performing spatial, temporal, and contextual reasoning by correlating observations from multiple heterogeneous data sources. • Generating explainable, evidence-based recommendations supported by maps, charts, geospatial visualizations, and marine advisories. • Enhancing fishermen safety through proactive alerts for adverse weather, high waves, lightning, cyclones, and other hazardous marine conditions. • Providing geofencing-based notifications when approaching international maritime boundaries, restricted waters, marine protected areas, ecologically sensitive zones, or other predefined operational boundaries. • Assisting with route optimization, safe navigation, and operational planning based on prevailing and forecast marine conditions. • Delivering reliable recommendations together with the supporting evidence and reasoning used to derive each response. Participants are encouraged to design a modular multi-agent architecture comprising specialized AI agents for planning, marine data discovery, weather intelligence, ocean analytics, geospatial reasoning, risk assessment, visualization, reporting, and user interaction. The architecture should demonstrate autonomous collaboration among agents to solve complex marine intelligence problems while providing an intuitive conversational experience.

SIH26178 HARDWARE | Qualcomm Inc

A resilient, AI-powered environmental monitoring network that provides early detection, localized intelligence, and actionable alerts for floods, forest fires, pollution events, and other environmental hazards common in India, enabling authorities and communities to shift from reactive disaster response to proactive risk prevention.

Theme: Space Technology **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Background India faces a growing range of environmental and climate-related risks including urban flooding, river floods, cyclones, forest fires, air pollution, droughts, landslides, and extreme weather events. Floods remain among the most frequent disasters across states such as Assam, Bihar, Kerala, and Maharashtra, while forest fires increasingly affect Uttarakhand, Himachal Pradesh, and central Indian forests. Air pollution continues to impact major urban centers, and climate change is increasing the frequency and intensity of these hazards. Government agencies such as NDMA, IMD, and ISRO already rely on environmental monitoring and early warning systems to support disaster management. Traditional monitoring systems often depend on centralized infrastructure and may not provide sufficiently localized, real-time intelligence. A distributed network of smart sensors powered by edge AI can improve early detection, reduce response times, and enable communities to act before environmental risks escalate into disasters. Description Design an Environmental Intelligence Network, a distributed system of interconnected AI-powered sensor nodes deployable across cities, rivers, forests, industrial zones, and vulnerable communities. Each node should use local (on device) AI inference to continuously monitor environmental conditions and identify emerging risks such as: • Rising water levels and flash flooding • Forest fires and smoke events • Hazardous air pollution • Extreme heat conditions • Landslide precursors • Industrial emissions or chemical leaks • Water quality degradation The sensor network should process data locally to reduce latency, minimize bandwidth requirements, and continue operating even during network outages. Only critical alerts, summarized insights, and risk assessments should be transmitted to regional control centers or disaster management authorities. Edge AI approaches enable devices to operate effectively in low-connectivity environments while providing rapid detection and decision support. Expected Solution The proposed solution should include: 1. Distributed Smart Sensor Nodes • Environmental sensors for water level, rainfall, temperature, humidity,smoke, air quality (PM2.5/PM10), gas leakage, soil moisture, and vibration. • Solar-powered, low-maintenance deployments suitable for remote locations. 2. On-Device AI Analytics • Real-time anomaly detection at the edge. • AI models capable of identifying flood risk, wildfire indicators, air-quality deterioration, and landslide warning signs. • Operation without continuous cloud connectivity. 3. Multi-Hazard Early Warning System • Automated alerts for: o Flooding and flash floods o Forest fires o Hazardous pollution episodes o Extreme weather conditions o Industrial safety incidents 4. Regional Environmental Risk Mapping • Geospatial visualization of sensor data. • Dynamic risk maps showing hotspots, risk trends, and affected zones. • Integration with emergency management dashboards. 5. Community and Authority Notification • Mobile and web alerts for local authorities and citizens. • Prioritized warning levels based on severity and confidence scores. 6. Cloud and Edge Hybrid Architecture • Edge processing for immediate decisions. • Centralized analytics for long-term trend analysis, forecasting, and policy support. 7. Scalable and Cost-Effective Deployment • Modular architecture that can scale from a single village to a smart city or state-wide deployment. • Support for IoT protocols such as LoRaWAN, NB-IoT, Wi-Fi, or 5G.

SIH26209 SOFTWARE | AICTE

Student Innovation

Theme: Space Technology **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Space technology refers to the application of engineering principles to the design, development, manufacture, and operation of devices and systems for space travel and exploration.

SIH26226 HARDWARE | AICTE

Student Innovation

Theme: Space Technology **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Space technology refers to the application of engineering principles to the design, development, manufacture, and operation of devices and systems for space travel and exploration.

🎯 THEME: MEDTECH / BIOTECH / HEALTHTECH

TOTAL: 18 SW: 13 HW: 5

SIH26018 SOFTWARE

Ministry of Rural Development

Intelligent Land Record Digitization and Validation System

Theme: MedTech / BioTech / HealthTech **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Intelligent Land Record Digitization and Validation System. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26033 SOFTWARE

Ministry of Consumer Affairs, Food & Public Distribution

Multiple intermediaries reduce farmers earnings and increase consumer prices.

Theme: MedTech / BioTech / HealthTech **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Multiple intermediaries reduce farmers earnings and increase consumer prices.. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26035 SOFTWARE

Ministry of Consumer Affairs, Food & Public Distribution

Development of a Software Program/Application for Generation of Test Reports for Non-Automatic Weighing Instruments (NAWI) as per OIML Recommendation R- 76

Theme: MedTech / BioTech / HealthTech **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

BACKGROUND:

Non-Automatic Weighing Instruments (NAWIs), such as electronic weighing scales, platform scales and weighbridges, are widely used in trade, commerce, healthcare, agriculture and industry where accurate measurement is essential for fair transactions and consumer protection.Under the Legal Metrology Act, 2009 and the Legal Metrology (General) Rules, 2011, such instruments used for transaction and protection are required to conform to prescribed standards,obtain model approval, and undergo verification and stamping before being put into use. For granting model approval, NAWIs are evaluated by designated laboratories in accordance with OIML Recommendation R 76 â€“ Non-Automatic Weighing Instruments, which specifies internationally accepted technical, metrological and performance requirements. The evaluation involves various metrological and functional tests, the results of which are compiled into detailed test reports.At present, these test reports are largely prepared manually using spreadsheets or document templates, making the process time-consuming, prone to calculation errors and lacking uniformity. Hence, there is a requirement to develop a software application that automates test data recording, compliance evaluation and generation of standardized test reports as per OIML R 76, thereby improving accuracy, consistency, and efficiency in the model approval process.

DESCRIPTION:

Develop a software application capable of generating complete test reports for Non-Automatic Weighing Instruments based on test observations recorded during type evaluation as per OIML R 76. The system should be capable of: - Capturing instrument details and technical specifications. • Recording laboratory and environmental conditions. • Entering observations from various OIML R 76 test procedures. • Automatically calculating permissible errors, and compliance status. • Performing validation checks for entered test data. • Automatically determining pass/fail criteria based on OIML R 76 requirements. • Generating standardized digital test reports in printable formats. • Maintaining a digital repository of completed test reports. • Providing secure user access with role-based permissions. • Supporting future updates whenever OIML recommendations are revised.

EXPECTED SOLUTION:

The proposed solution should include: • User-friendly desktop and/or web-based application. • Digital data entry forms for all applicable OIML R 76 tests. • Automated calculations and compliance verification. • Standardized test report generation in PDF and editable formats MS Word etc. • Instrument-wise test history and report repository. • Dashboard for monitoring testing activities and report status. • Search and retrieval facility for previously generated reports. • Technical documentation describing software architecture, calculation methodology and deployment framework. Key Functional Requirements: • Entry of manufacturer details, Instrument specifications, Model information and technical parameters • Compliance determination as per OIML R-76 • Entry of observations for all prescribed tests • Automatic validation of input data and related calculations • Automatic preparation of standardized test reports with auto-population of laboratory and instrument details • Attachment of photographs and supporting documents • Digital signatures (optional) • Export to PDF and editable formats • Dashboard for test report management (completed, in process, history access etc.)

SIH26043 SOFTWARE

Governmcnt of Jharkhand

A digital platform to crowdsource societal challenges and facilitate collaborative problem solving through universities and industry partnerships

Theme: MedTech / BioTech / HealthTech **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

BACKGROUND:

Communities across Jharkhand encounter numerous local challenges related to education,healthcare, agriculture, water management, sanitation, environment, rural livelihoods,accessibility, urban infrastructure, and public service delivery. While citizens are often the first to identify these issues, there is currently no structured mechanism through which they can submit such problems for systematic evaluation and innovation-driven resolution.At the same time, Higher Education Institutions (HEIs) possess significant academic expertise,research capabilities, and a large pool of students capable of developing practical solutions.Industries and start-ups also have technical expertise, financial resources, and implementation capabilities that can complement academic research. However, collaboration among citizens,universities, and industry remains largely fragmented and project-specific.The National Education Policy (NEP) 2020 emphasizes experiential learning, multidisciplinary research, innovation, industry collaboration, and community engagement. Establishing a technology-enabled platform that connects societal challenges with academic institutions and industry partners can foster demand-driven innovation while enabling students and researchers to work on real-world problems that create measurable social impact.

DESCRIPTION:

Every year, citizens across Jharkhand identiff thousands of local issues that require innovative technological or process-based solutions. These challenges often remain unresolved due to the absence of a centralized platform that enables problem collection, categorization, expert evaluation, institutional assignment, and industry collaboration.There is a need to develop a digital platform capable of: • Allowing citizens, community organizations, local bodies, and government agencies to submit societal challenges thiough an intuitive web and mobile interface, supported by photographs, videos, location details, and relevant documents. • Automatically categorizing submitted problems based on thematic domains such as education, agriculture, healthcare, water resources, environment, energy, urban development, accessibility, public administration, and rural livelihoods using AI-enabled classification techniques. • Routing validated problem statements to appropriate universities based on their academic disciplines, research expertise, innovation centres, incubation facilities, and faculty specialization. • Enabling universities to evaluate submitted challenges, constitute multidisciplinary student and faculty teams, and prepare solution proposals or research projects. • Facilitating collaboration between universities and industry partners, startups, MSMEs,CSR organizations, research laboratories, and innovation ecosystems for mentorship,funding, prototyping, testing and deployment of solutions. • Providing workflow management for problem review, institutional allocation, project monitoring, stakeholder communication, milestone tracking, and solution validation. • Generating dashboards and analytics for govemment departments to monitor the number of challenges received, domain-wise distribution, institutional participation, industry engagement, project progress, and measurable social outcomes. The platform should support a transparent and scalable innovation ecosystem that transforms community-driven challenges into research, innovation, entrepreneurship, and deployable solutions.

EXPECTED SOLUTION:

A comprehensive Societal Innovation Collaboration Portal comprising the following components: • A citizen engagement module enabling individuals, community groups, Panchayati Raj Institutions, Urban Local Bodies, and government departments to submit societal challenges with multimedia evidence, geographical location, and supporting information. • An AI-enabled problem management module capable of automatically categorizing, prioritizing, deduplication, and routing validated challenges to appropriate universities based on subject expertise and institutional capabilities. • A university collaboration module allowing Higher Education Institutions to review assigned challenges, form multidisciplinary project teams, assign faculty mentors, manage project workflows, and submit solution proposals. • An industry partnership module facilitating participation by industries, startups, MSMEs,CSR organizations, research institutions, and innovation hubs for mentoring, co-development, funding, prototyping, pilot implementation, and technology transfer. • A project lifecycle management system for monitoring milestones, deliverables, approvals, documentation, testing outcomes, intellectual property generation, and implementation status. • A visual analytics dashboard providing real-time insights on challenge submissions, university participation, industry collaborations, thematic trends, project completion rates,innovation outcomes, patents, startups created, and community impact across districts and sectors. • A notification and communication system enabling seamless interaction among citizens,universities, industry partners, mentors, and government departments throughout the project lifecycle.

SIH26050 HARDWARE DRDO

High Altitude Performance Optimization and Robust Design of Anti-Drone System.

Theme: MedTech / BioTech / HealthTech **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: High Altitude Performance Optimization and Robust Design of Anti-Drone System.. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26076 SOFTWARE Ministry of Earth Sciences (MoES)

Development of personalized homepage for 'Mausam' mobile application:

Theme: MedTech / BioTech / HealthTech **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

• Health-conscious users Highlight Air Quality Index (AQI), pollen count, UV index, and humidity levels to help users manage allergies, asthma, or skin sensitivity. • Outdoor fitness enthusiasts Show sunrise/sunset times, 'best running hours,' wind speed, and heat alerts to optimize workout planning. • Beachgoers & surfers Display sea conditions, tide timings, wave height, and water temperature for safe and enjoyable beach activities. • Travelers Provide quick access to saved destinations, severe weather alerts for flights, and packing suggestions (e.g., 'Carry a raincoat in London'). • Parents & families Emphasize school commute conditions, rain alerts, and severe weather warnings to plan daily routines. • Agriculture & gardeners Show soil moisture, rainfall predictions, frost alerts, and seasonal planting guidance. • Commuters Integrate weather with traffic updates, visibility conditions, and alerts for storms or fog that affect travel. • Event planners Offer extended forecasts, probability of rain, and 'comfort index' for outdoor gatherings or weddings.

SIH26094 SOFTWARE Ministry of Social Justice and Empowerment (MoSJE)

AI-Powered Dynamic Mental Health Monitoring and Distress Prediction System for Victims of Atrocities

Theme: MedTech / BioTech / HealthTech **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AI-Powered Dynamic Mental Health Monitoring and Distress Prediction System for Victims of Atrocities. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26113 HARDWARE Autodesk

Human augmentation technologies are transforming healthcare,rehabilitation, industrial ergonomics, assistive living, sports, and personal mobility by improving human capabilities and enhancing quality of life.

Theme: MedTech / BioTech / HealthTech **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Human augmentation technologies are transforming healthcare,rehabilitation, industrial ergonomics, assistive living, sports, and personal mobility by improving human capabilities and enhancing quality of life.. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26115 SOFTWARE Autodesk

Design and Develop a Smart Mobile Medical-Waste Collection and Segregation System

Theme: MedTech / BioTech / HealthTech **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Design and Develop a Smart Mobile Medical-Waste Collection and Segregation System. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26131 SOFTWARE Government Of Maharashtra

Early detection and management of crop diseases and pest infestations

Theme: MedTech / BioTech / HealthTech **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

• Problem Description Farmers often recognise crop diseases or pest infestations only after visible damage has spread. Extension staff may cover large areas, while laboratory diagnosis and expert advice may not be immediately available. Weather, crop stage, variety, soil condition and local pest history influence risk, but these inputs are rarely combined into actionable farm-level alerts. Incorrect diagnosis may lead to delayed treatment, excessive or inappropriate pesticide use, increased cultivation cost,residue concerns and yield loss. The challenge is to provide timely, reliable and locally relevant detection,forecasting and management support. • Expected Solution / Outcome A farmer- and extension-worker-friendly crop-health system that supports image based symptom identification, pest-trap or sensor inputs, weather-based risk forecasting, geospatial hotspot mapping,expert validation and multilingual advisories. The system should recommend integrated pest and disease management actions, safe input usage,referral to extension or laboratories, and follow-up monitoring. It should learn from field confirmations and provide dashboards for agriculture officials.Expected outcomes include earlier detection, reduced crop loss, more targeted pesticide use, faster extension response, improved surveillance coverage and better planning of preventive interventions.

SIH26133 SOFTWARE Government Of Maharashtra

Accessibility and quality of public healthcare services,particularly in rural and underserved areas

Theme: MedTech / BioTech / HealthTech **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Accessibility and quality of public healthcare services,particularly in rural and underserved areas. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26139 SOFTWARE Egreen Quanta

Hybrid Quantum Machine Learning Platform for Early Disease Detection

Theme: MedTech / BioTech / HealthTech **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Hybrid Quantum Machine Learning Platform for Early Disease Detection. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26181 HARDWARE Qualcomm Inc

A secure, AI-powered Personal Health Companion that delivers real-time, privacy-preserving health monitoring and early warning capabilities, helping individuals recognize health risks before they become emergencies. The solution should improve resilience during heat waves, floods, pollution events, and other disasters common in India while enabling continuous health support through on-device intelligence.

Theme: MedTech / BioTech / HealthTech **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: A secure, AI-powered Personal Health Companion that delivers real-time, privacy-preserving health monitoring and early warning capabilities, helping individuals recognize health risks before they become emergencies. The solution should improve resilience during heat waves, floods, pollution events, and other disasters common in India while enabling continuous health support through on-device intelligence.. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26186 SOFTWARE

Ministry of Home Affairs

AI-Based Predictive Personnel Stress and Welfare Monitoring System for Uniformed Forces

Theme: MedTech / BioTech / HealthTech **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AI-Based Predictive Personnel Stress and Welfare Monitoring System for Uniformed Forces. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26196 SOFTWARE

AICTE

Student Innovation

Theme: MedTech / BioTech / HealthTech **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Student Innovation. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26200 SOFTWARE

AICTE

Student Innovation

Theme: MedTech / BioTech / HealthTech **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Student Innovation. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26213 HARDWARE

AICTE

Student Innovation

Theme: MedTech / BioTech / HealthTech **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Student Innovation. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26217 HARDWARE

AICTE

Student Innovation

Theme: MedTech / BioTech / HealthTech **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Student Innovation. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

🎯 THEME: SMART EDUCATION

TOTAL: 17 SW: 12 HW: 5

SIH26042 SOFTWARE Governmcnt of Jharkhand

AI-Powered Vernacular Pedagogy and Real-Time Translation Tool for Mother Tongue-Based Primary Education

Theme: Smart Education **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AI-Powered Vernacular Pedagogy and Real-Time Translation Tool for Mother Tongue-Based Primary Education. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26059 SOFTWARE Ministry of Earth Sciences (MoES)

AI-Enabled Antarctic Sea-Ice, Iceberg Trajectory, and Navigation Decision Support System

Theme: Smart Education **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AI-Enabled Antarctic Sea-Ice, Iceberg Trajectory, and Navigation Decision Support System. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26070 SOFTWARE Ministry of Earth Sciences (MoES)

To develop an Artificial Intelligence (AI) / Machine Learning (ML) based system for identification, classification, and prediction of different tropical cyclone patterns using multi-source satellite data.

Theme: Smart Education **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: To develop an Artificial Intelligence (AI) / Machine Learning (ML) based system for identification, classification, and prediction of different tropical cyclone patterns using multi-source satellite data.. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26075 SOFTWARE Ministry of Earth Sciences (MoES)

Participants are invited to design and develop **CAPACITY CONNECT A Digital Capacity Building and Learning Management Portal** to support organizational training, competency development, and knowledge sharing through a centralized web-based platform.

Theme: Smart Education **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Participants are invited to design and develop **CAPACITY CONNECT A Digital Capacity Building and Learning Management Portal** to support organizational training, competency development, and knowledge sharing through a centralized web-based platform.. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26087 HARDWARE Ministry of Cooperation

AI-Enabled Cooperative Capacity Building, ERP & Employment Ecosystem

Theme: Smart Education **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AI-Enabled Cooperative Capacity Building, ERP & Employment Ecosystem. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26097 SOFTWARE

Ministry of Social Justice and Empowerment (MoSJE)

AI-Driven voice Assistant for livelihood Mapping and NSQF-Aligned Skilling Recommendations for SC Communities under GIA component of PM-AJAY

Theme: Smart Education **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AI-Driven voice Assistant for livelihood Mapping and NSQF-Aligned Skilling Recommendations for SC Communities under GIA component of PM-AJAY. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26116 SOFTWARE

Autodesk

Urban Mixed-Use Design Challenge-Design a centrally located mixed-use building in Autodesk Revit with commercial spaces (Ground + 1st floor) and residential units (up to 8 floors). 1 Level of Basement (Car Parking + EV Charging), Total (B+G+9)(Note: Plot size and all required dimensions may be assumed by students (in mm units).

Theme: Smart Education **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Urban Mixed-Use Design Challenge-Design a centrally located mixed-use building in Autodesk Revit with commercial spaces (Ground + 1st floor) and residential units (up to 8 floors). 1 Level of Basement (Car Parking + EV Charging), Total (B+G+9)(Note: Plot size and all required dimensions may be assumed by students (in mm units).. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26135 SOFTWARE

Government Of Maharashtra

Difficulties in tracking employment outcomes,skill gaps, and the impact of skilling initiatives

Theme: Smart Education **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Difficulties in tracking employment outcomes,skill gaps, and the impact of skilling initiatives. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26140 SOFTWARE

Egreen Quanta

AI-Based Interactive Quantum Algorithm Learning Platform

Theme: Smart Education **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AI-Based Interactive Quantum Algorithm Learning Platform. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26202 SOFTWARE

AICTE

Student Innovation

Theme: Smart Education **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

A solution/idea that can boost the current situation of the tourism industries including hotels, travel and others.

SIH26204 SOFTWARE AICTE

Student Innovation

Theme: Smart Education **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Student Innovation. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26205 SOFTWARE AICTE

Student Innovation

Theme: Smart Education **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Smart education, a concept that describes learning in digital age. It enables learners to learn more effectively, efficiently, flexibly and comfortably.

SIH26208 SOFTWARE AICTE

Student Innovation

Theme: Smart Education **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Challenge your creative mind to conceptualize and develop unique toys and games based on our civilization, history, and culture etc.

SIH26219 HARDWARE AICTE

Student Innovation

Theme: Smart Education **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

A solution/idea that can boost the current situation of the tourism industries including hotels, travel and others.

SIH26221 HARDWARE AICTE

Student Innovation

Theme: Smart Education **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Student Innovation. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26222 HARDWARE AICTE

Student Innovation

Theme: Smart Education **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Smart education, a concept that describes learning in digital age. It enables learners to learn more effectively, efficiently, flexibly and comfortably.

SIH26225 HARDWARE

AICTE

Student Innovation

Theme: Smart Education **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Challenge your creative mind to conceptualize and develop unique toys and games based on our civilization, history, and culture etc.

🎯 THEME: AGRICULTURE, FOODTECH & RURAL DEVELOPMENT

TOTAL: 15 SW: 11 HW: 4

SIH26016 SOFTWARE

Ministry of Rural Development

Real-Time National Land Acquisition & Management System for End-to-End Digital Monitoring and Decision Support

Theme: Agriculture, FoodTech & Rural Development **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

BACKGROUND:

Land acquisition is a critical component of infrastructure development and public welfare projects in India. It facilitates the implementation of highways, railways, industrial corridors, irrigation projects, urban development, renewable energy initiatives, and other strategic infrastructure. The process involves multiple stakeholders, including land requiring bodies, land acquiring authorities, district administrations, state governments, and central ministries. Description of the Study: Web-based National Land Acquisition & Management System that digitizes the complete land acquisition lifecycleâ€"from project proposal submission to final possession of land. The proposed platform should provide standardized workflows for different stakeholders and enable seamless coordination among Central Ministries, State Governments, District Authorities, and Project Implementing Agencies. The system should facilitate online submission and approval of proposals, digital scrutiny, document management, automated workflow routing, and status tracking at every stage.The platform should support geo-tagging of acquired land parcels using GIS technology, enabling visualization of project locations on interactive maps. It should also maintain real-time information on key land acquisition parameters such as:

- **7.** Land proposed and acquired 8. Notifications issued 9. Awards declared 10. Compensation assessed and disbursed 11. Possession status 12. Rehabilitation and Resettlement (R&R) progress 13. Number of affected and displaced families 14. Project-wise and state-wise progress 15. Timeline monitoring and milestone tracking The platform should be scalable for nationwide implementation across all States and Union Territories while ensuring data security, interoperability, and compliance with government standards. Add 'Scope of Study' Table here Problems: At present, land acquisition activities are managed through fragmented systems, manual documentation, and state-specific processes. The absence of a unified national digital platform results in inconsistent data collection, duplication of efforts, delays in approvals, limited transparency, and inadequate monitoring. Decision-makers often lack access to real-time information on the progress of land acquisition, compensation disbursement, possession status, and rehabilitation measures.

EXPECTED SOLUTION:

The solution should incorporate role-based access control, automated alerts and notifications, API-based integration with relevant government systems, customizable dashboards, analytical reports, and predictive analytics to support policy formulation and efficient project execution. The system should provide:

- **7.** End-to-end digital workflow for land acquisition processes; 8. Online submission, verification, approval, and tracking of proposals; 9. GIS-enabled geo-tagging and spatial visualization of land parcels; 10. Interactive national dashboard displaying: Area notified, Area acquired, Compensation assessed and paid, Number of affected and displaced families, Rehabilitation & Resettlement status, Project progress, Possession status, Timeline adherence 11. API-based integration with land records, cadastral maps, and relevant government portals.
- **12.** Mobile-responsive interface for field-level data collection and verification.
- **13.** Secure document repository with version control and audit history.
- **14.** Customizable MIS reports and executive dashboards for decision-makers. The solution should significantly enhance transparency, accountability, efficiency, and data-driven governance in land acquisition while reducing processing time and improving inter-agency coordination. Add 'Suggested components-wise technology' table here

SIH26017 SOFTWARE

Ministry of Rural Development

Predictive Analytics System for Early Detection of Land Acquisition Delays

Theme: Agriculture, FoodTech & Rural Development **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Predictive Analytics System for Early Detection of Land Acquisition Delays. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26022 HARDWARE Ministry of MSME

Design and develop a smart, solar-powered drying and compact packaging system to support home-based agarbatti manufacturing by rural women artisans.

Theme: Agriculture, FoodTech & Rural Development **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Design and develop a smart, solar-powered drying and compact packaging system to support home-based agarbatti manufacturing by rural women artisans.. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26034 SOFTWARE Ministry of Consumer Affairs, Food & Public Distribution

Software System to check compliance of Packaged Commodities under Legal Metrology(Packaged Commodities) Rules, 2011 by scanning products, images and labels.

Theme: Agriculture, FoodTech & Rural Development **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Software System to check compliance of Packaged Commodities under Legal Metrology(Packaged Commodities) Rules, 2011 by scanning products, images and labels.. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26051 SOFTWARE DRDO

Software Based Model Development for Design of Area Specific Shelter for Thermal Comfort Maintenance.

Theme: Agriculture, FoodTech & Rural Development **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Software Based Model Development for Design of Area Specific Shelter for Thermal Comfort Maintenance.. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26086 SOFTWARE Ministry of Earth Sciences (MoES)

Hyperlocal Monsoon Onset & Break Prediction System (Block/Village Scale)

Theme: Agriculture, FoodTech & Rural Development **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

The Indian Summer Monsoon dictates the economic livelihood of millions of farmers, particularly during the Kharif sowing season. While macro-scale monsoon forecasts across large meteorological subdivisions have improved, Indian agriculture remains highly vulnerable to the unpredictable nature of intra-seasonal variations. Specifically, the exact dates of monsoon onset,prolonged dry spells (break-monsoon phases), and subsequent revival cycles vary drastically from one district to another. Standard regional forecasts lack the spatial granularity required for localized agricultural planning.If a farmer sows seeds during a false onset just before a major breakthrough pause, entire crops fail due to moisture stress, leading to crushing financial losses. The challenge is to build a hybrid predictive framework capable of delivering a 7-to-30-day probabilistic outlook of monsoon behavior at the Block and Panchayat (Village cluster) scale. The system must bridge the gap between global climate teleconnections and hyper-local weather outcomes. Participants should design a solution that ingests large-scale climate indicesâ€”such as the El NiÃ±o-Southern Oscillation (ENSO), Indian Ocean Dipole (IOD), and Madden-Julian Oscillation (MJO)â€”and downscales their signatures using advanced machine learning models to predict localized precipitation behavior, onset thresholds, and active/break durations.Develop a hybrid mathematical or machine learning model that pairs global planetary boundary conditions (ENSO, IOD, MJO phases) with regional atmospheric data to predict local rainfall anomalies. Generate dynamic, color-coded risk maps at the block/panchayat level illustrating the statistical probability percentage of monsoon onset, continuous dry spells (breaks), or heavy downpours 1 to 4 weeks in advance. Build an expert-system engine that translates rainfall probabilities into localized crop-specific agronomic advisories (e.g., advising farmers to delay sowing, prepare irrigation alternatives, or alter crop choices based on upcoming break phases). A mobile-optimized web application or automated SMS/WhatsApp API gateway that pushes clear,actionable text-based advisories in regional Indian languages directly to farmers and local agricultural extension officers.

SIH26091 SOFTWARE Ministry of Social Justice and Empowerment (MoSJE)

AI-Driven Hyper-Local Business Advisory and Financial Structuring Assistant for Rural Micro-Entrepreneurs

Theme: Agriculture, FoodTech & Rural Development **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AI-Driven Hyper-Local Business Advisory and Financial Structuring Assistant for Rural Micro-Entrepreneurs. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26109 HARDWARE Ministry of Fisheries, Animal Husbandry & Dairying

AI-Based Predictive Modelling for Early Forecasting of Bovine Mastitis in Indian Dairy Farms

Theme: Agriculture, FoodTech & Rural Development **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AI-Based Predictive Modelling for Early Forecasting of Bovine Mastitis in Indian Dairy Farms. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26110 HARDWARE Ministry of Fisheries, Animal Husbandry & Dairying

Development of a Low-Cost Light-weight Milk Chilling Can for Small-Scale Dairy Farmers

Theme: Agriculture, FoodTech & Rural Development **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Development of a Low-Cost Light-weight Milk Chilling Can for Small-Scale Dairy Farmers. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26111 SOFTWARE Ministry of Fisheries, Animal Husbandry & Dairying

Smart AI-Enabled Rapid Feed and Silage Quality Testing System for Dairy Farmers

Theme: Agriculture, FoodTech & Rural Development **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Smart AI-Enabled Rapid Feed and Silage Quality Testing System for Dairy Farmers. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26126 SOFTWARE Bharat Electronics Limited

Vision Based Autonomous Navigation for Unmanned Ground Vehicle for Outdoor environment

Theme: Agriculture, FoodTech & Rural Development **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

• Background Outdoor Unmanned Ground Vehicles (UGVs) face unpredictable terrain, changing light, and unreliable GPS signals. To achieve true autonomy in applications like search-and-rescue,agriculture, or delivery, UGVs must rely on onboard computer vision. Visual perception provides a cost-effective, data-rich way for vehicles to understand and safely navigate complex,unstructured outdoor surroundings. • Description The objective is to build an autonomous navigation system for a UGV operating in a GPS-denied outdoor environment using camera feeds as the primary sensor. Students must solve three key challenges: 1. Path Detection: Real-time identification of safe, traversable paths vs. hazards (e.g., rocks,ditches, trees). 2. Visual Localization: Estimating the UGVâ€™s position and orientation without GPS using visual data. 3. Collision Avoidance: Dynamically routing the vehicle around sudden obstacles toward a destination. • Expected Solution A functional software module consisting of: • Perception AI: A lightweight model for obstacle and path detection. • Visual SLAM/Odometry: A pipeline to track vehicle movement. • Path Planner: An algorithm to translate visual data into wheel/motor commands. • Success Criteria: Successful, collision-free navigation from Point A to Point B across outdoor scenarios

SIH26128 SOFTWARE Government Of Maharashtra

Efficient systems for early detection,prevention,and management of livestock diseases and animal health issues

Theme: Agriculture, FoodTech & Rural Development **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Efficient systems for early detection,prevention,and management of livestock diseases and animal health issues. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26132 SOFTWARE Government Of Maharashtra

Strengthening market linkages and price discovery for farmers

Theme: Agriculture, FoodTech & Rural Development **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Strengthening market linkages and price discovery for farmers. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26197 SOFTWARE AICTE

Student Innovation

Theme: Agriculture, FoodTech & Rural Development **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Developing solutions, keeping in mind the need to enhance the primary sector of India - Agriculture and to manage and process our agriculture produce.

SIH26214 HARDWARE AICTE

Student Innovation

Theme: Agriculture, FoodTech & Rural Development **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Developing solutions, keeping in mind the need to enhance the primary sector of India - Agriculture and to manage and process our agriculture produce.

🎯 THEME: ROBOTICS AND DRONES

TOTAL: 8 SW: 6 HW: 2

SIH26012 SOFTWARE Ministry of Rural Development

AI-Based Automated Urban Parcel Mapping and Cadastral Feature Extraction System using Drone Imagery

Theme: Robotics and Drones **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AI-Based Automated Urban Parcel Mapping and Cadastral Feature Extraction System using Drone Imagery. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26014 SOFTWARE Ministry of Rural Development

An Integrated GIS-based Digital Public Infrastructure for Land Governance

Theme: Robotics and Drones **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: An Integrated GIS-based Digital Public Infrastructure for Land Governance. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26026 HARDWARE Ministry of Railways

Development of Mobile (Quadruped)/Handheld Device/System for Real-Time Detection of Narcotics and Explosives across Indian Railways.

Theme: Robotics and Drones **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Development of Mobile (Quadruped)/Handheld Device/System for Real-Time Detection of Narcotics and Explosives across Indian Railways.. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26037 SOFTWARE MathWorks

Adaptive Path Planning and Collision Avoidance for Autonomous Vehicles on Unstructured Indian Roads

Theme: Robotics and Drones **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Adaptive Path Planning and Collision Avoidance for Autonomous Vehicles on Unstructured Indian Roads. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26054 SOFTWARE DRDO

AI-Enabled Real-Time Digital Twin System for Health Monitoring, Fault Prediction and Mission Reliability Enhancement of Aero Piston Engines used in MALE UAVs.

Theme: Robotics and Drones **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AI-Enabled Real-Time Digital Twin System for Health Monitoring, Fault Prediction and Mission Reliability Enhancement of Aero Piston Engines used in MALE UAVs.. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26112 HARDWARE Autodesk

Design and Develop a Modular Autonomous Mobile Robot (AMR) Platform for Smart Warehouse Automation

Theme: Robotics and Drones **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Design and Develop a Modular Autonomous Mobile Robot (AMR) Platform for Smart Warehouse Automation. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26123 SOFTWARE | Bharat Electronics Limited

Edge-AI Based Distributed Fleet Coordination for Autonomous Mobile Robots (AMRs) in Smart Warehouses

Theme: Robotics and Drones **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Edge-AI Based Distributed Fleet Coordination for Autonomous Mobile Robots (AMRs) in Smart Warehouses. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26158 SOFTWARE | National Technical Research Organisation (NTRO)

Single-Pass Drone Video to Accurate 3D Model Generation System

Theme: Robotics and Drones **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Single-Pass Drone Video to Accurate 3D Model Generation System. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

🎯 THEME: FITNESS & SPORTS

TOTAL: 8 SW: 5 HW: 3

SIH26031 SOFTWARE | Ministry of Consumer Affairs, Food & Public Distribution

Quality assessment and grading of onions are often subjective and vary across procurement centers, resulting in disputes and inconsistencies.

Theme: Fitness & Sports **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Quality assessment and grading of onions are often subjective and vary across procurement centers, resulting in disputes and inconsistencies.. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26048 HARDWARE | Ministry of Ayush

iKwath - a pod-based smart Kwatha (Kadha) maker that prepares a fresh, AFI/API-standardized decoction from coarse powder (yavaku?a c?r?a) on demand, in the shortest practical time without altering the decoctions quality or yield

Theme: Fitness & Sports **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: iKwath - a pod-based smart Kwatha (Kadha) maker that prepares a fresh, AFI/API-standardized decoction from coarse powder (yavaku?a c?r?a) on demand, in the shortest practical time without altering the decoctions quality or yield. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26124 SOFTWARE | Bharat Electronics Limited

AI-Powered Mobile Urban Intelligence Platform Using Public Transport Fleet

Theme: Fitness & Sports **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AI-Powered Mobile Urban Intelligence Platform Using Public Transport Fleet. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26137 SOFTWARE

Egreen Quanta

Quantum-Inspired Intelligent Traffic Route Optimization in Transportation Systems Using Metaheuristic Optimization

Theme: Fitness & Sports **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Quantum-Inspired Intelligent Traffic Route Optimization in Transportation Systems Using Metaheuristic Optimization. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26194 SOFTWARE

AICTE

Student Innovation

Theme: Fitness & Sports **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Student Innovation. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26199 SOFTWARE

AICTE

Student Innovation

Theme: Fitness & Sports **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Student Innovation. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26211 HARDWARE

AICTE

Student Innovation

Theme: Fitness & Sports **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Student Innovation. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26216 HARDWARE

AICTE

Student Innovation

Theme: Fitness & Sports **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Student Innovation. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

🎯 THEME: TRAVEL & TOURISM

TOTAL: 4 SW: 2 HW: 2

SIH26039 HARDWARE

Governmcnt of Jharkhand

AI-Powered Underground Mine Safety, Monitoring and Rescue System.

Theme: Travel & Tourism **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: AI-Powered Underground Mine Safety, Monitoring and Rescue System.. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26056 SOFTWARE

MoSPI

Development of a Real-time Airfare Price Index for India through Automated Web Scraping of Airline and Online Travel Aggregator Portals for Augmentation of the Consumer Price Index (CPI).

Theme: Travel & Tourism **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Development of a Real-time Airfare Price Index for India through Automated Web Scraping of Airline and Online Travel Aggregator Portals for Augmentation of the Consumer Price Index (CPI).. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26207 SOFTWARE

AICTE

Student Innovation

Theme: Travel & Tourism **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Student Innovation. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26224 HARDWARE

AICTE

Student Innovation

Theme: Travel & Tourism **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Student Innovation. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

🎯 THEME: SMART VEHICLES

TOTAL: 3 SW: 1 HW: 2

SIH26005 HARDWARE

Ministry of Development of North Eastern Region (MDoNER)

Solar-Powered Smart Mini Cold Storage System for Fresh Vegetables in North Eastern Region (NER)

Theme: Smart Vehicles **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Solar-Powered Smart Mini Cold Storage System for Fresh Vegetables in North Eastern Region (NER). Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26052 HARDWARE DRDO

To develop an AI/ML-enabled adaptive noise cancellation (ANC) system that effectively suppresses stationary, non-stationary, and impulsive defence noises while maintaining high speech intelligibility and real-time performance on embedded hardware.

Theme: Smart Vehicles **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: To develop an AI/ML-enabled adaptive noise cancellation (ANC) system that effectively suppresses stationary, non-stationary, and impulsive defence noises while maintaining high speech intelligibility and real-time performance on embedded hardware.. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26138 SOFTWARE Egreen Quanta

Quantum-Inspired Fuel Consumption Prediction and Green Fleet Optimization

Theme: Smart Vehicles **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Quantum-Inspired Fuel Consumption Prediction and Green Fleet Optimization. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

🎯 THEME: CLEAN & GREEN TECHNOLOGY

TOTAL: 2 SW: 2 HW: 0

SIH26038 SOFTWARE MathWorks

Explainable AI for Diabetic Retinopathy Screening in Rural India

Theme: Clean & Green Technology **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Explainable AI for Diabetic Retinopathy Screening in Rural India. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26055 SOFTWARE DRDO

Smart Scan strategy for Electronic Warfare

Theme: Clean & Green Technology **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Smart Scan strategy for Electronic Warfare. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

🎯 THEME: TOYS & GAMES

TOTAL: 2 | SW: 2 | HW: 0

SIH26045 SOFTWARE

Ministry of Ayush

IP-SAKTI Sahayak a multilingual, RAG-based (source-cited) AI assistant for Intellectual Property and regulatory guidance in Ayurveda, across national and international regimes.

Theme: Toys & Games **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: IP-SAKTI Sahayak a multilingual, RAG-based (source-cited) AI assistant for Intellectual Property and regulatory guidance in Ayurveda, across national and international regimes.. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

SIH26062 SOFTWARE

Ministry of Earth Sciences (MoES)

Integrated Polar Expedition Logistics and Asset Management System

Theme: Toys & Games **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Overview: Integrated Polar Expedition Logistics and Asset Management System. Sponsoring ministry specifications, guidelines, and submission requirements released on the official portal sih.gov.in.

🎯 THEME: RENEWABLE ENERGY

TOTAL: 1 | SW: 1 | HW: 0

SIH26061 SOFTWARE

Ministry of Earth Sciences (MoES)

AI-Driven Smart Energy Management System for Polar Research Stations

Theme: Renewable Energy **Prize:** ₹1,00,000 INR **Deadline:** 20 September 2026 **Portal:** sih.gov.in

Develop an intelligent energy-management system using AI for load forecasting, renewable energy integration and fuel optimization under extreme polar conditions.